

Equations implemented in `numerics/mrgw_hermitian`

A reimplementaion guide for the Hermitian pole-space MR-*GW* code

September 22, 2026

Purpose and reading guide

This document writes out, in the order in which the code evaluates them, all equations implemented in the package `numerics/mrgw_hermitian` and in the three scripts `run_checks.py`, `example_h4.py` and `check_ozone.py`. Every formula is stated with the index conventions, normalizations, thresholds and sign choices actually used, so that the methods can be reimplemented without reading the code. The theory behind the equations is in `notes/mr-hedin.tex` (Objectives I and II), `notes/comparison.tex` and Wang, Fang, and Li, arXiv:2604.16013 (WFL below); equation numbers of the form (Sn) refer to the Supplemental Material of WFL. Section 10 maps each equation to the function that implements it.

Throughout, all quantities are real (real orbitals, real CI vectors), and $\dagger$ can be read as a transpose. Energies are in Hartree inside the code; the scripts convert to eV with $1 \text{ Ha} = 27.211386245988 \text{ eV}$.

1 Conventions and PySCF primitives

Spin orbitals. Spatial molecular orbitals $p_s = 0, \dots, n_{\text{mo}} - 1$ are ordered core, active, virtual (indices $0 \leq p_s < n_{\text{core}}$, $n_{\text{core}} \leq p_s < n_{\text{occ}}$, $p_s \geq n_{\text{occ}} = n_{\text{core}} + n_{\text{cas}}$). Spin orbitals are labelled

$$p = 2p_s + \sigma, \quad \sigma = 0 (\alpha), 1 (\beta), \quad n_{\text{so}} = 2n_{\text{mo}}. \quad (1)$$

The index sets $\mathcal{C}$ (core), $\mathcal{A}$ (active), $\mathcal{V}$ (virtual) and $\mathcal{I} = \mathcal{C} \cup \mathcal{V}$ (inactive) are the corresponding lists of spin orbitals; “local active” indices $x_\ell = 2(x_s - n_{\text{core}}) + \sigma \in \{0, \dots, n_{\text{act}} - 1\}$, $n_{\text{act}} = 2n_{\text{cas}}$, label the active spin orbitals inside the CAS routines. Symbols: P, Q inactive; i, j core; a, b virtual; x, y, z, w active; p, q, r, s arbitrary.

Integrals. With C the (canonicalized, Sec. 2.2) MO coefficients, $h^{\text{MO}} = C^T h^{\text{AO}} C$ and the chemist-notation two-electron integrals $(p_s r_s | q_s s_s)$ from `ao2mo.kernel` restored to a four-index array,

$$h_{pq} = h_{p_s q_s}^{\text{MO}} \delta_{\sigma_p \sigma_q}, \quad (2)$$

$$V[p, r, q, s] \equiv v_{pr,qs} = (p_s r_s | q_s s_s) \delta_{\sigma_p \sigma_r} \delta_{\sigma_q \sigma_s} = \langle pq | rs \rangle, \quad (3)$$

$$\bar{v}_{pr,qs} = v_{pr,qs} - v_{ps,qr} \quad (\mathbf{V} - \mathbf{V.transpose}(0,3,2,1)), \quad (4)$$

so that the interaction is $\frac{1}{2} \sum v_{pr,qs} a_p^\dagger a_q^\dagger a_s a_r = \frac{1}{4} \sum \bar{v}_{pr,qs} a_p^\dagger a_q^\dagger a_s a_r$ (WFL convention). The residual interaction removes the all-active elements,

$$v_{pr,qs}^R = (1 - \delta_{p \in \mathcal{A}} \delta_{r \in \mathcal{A}} \delta_{q \in \mathcal{A}} \delta_{s \in \mathcal{A}}) v_{pr,qs}, \quad \bar{v}_{pr,qs}^R = v_{pr,qs}^R - v_{ps,qr}^R. \quad (5)$$

CI vectors and operators. CI vectors of a sector with (n_α, n_β) electrons in n spatial orbitals are PySCF arrays of shape $(N_{n_\alpha}, N_{n_\beta})$, $N_k = \binom{n}{k}$ (`cistring.num_strings`). The elementary operations are

$$\hat{a}_p^\dagger |v\rangle = \text{cre_a}(v, n, (n_\alpha, n_\beta), p_s) \text{ if } \sigma_p = 0, \quad \text{cre_b}(v, n, (n_\alpha, n_\beta), p_s) \text{ if } \sigma_p = 1, \quad (6)$$

$$\hat{a}_p |v\rangle = \text{des_a}(v, n, (n_\alpha, n_\beta), p_s) \text{ if } \sigma_p = 0, \quad \text{des_b}(v, n, (n_\alpha, n_\beta), p_s) \text{ if } \sigma_p = 1, \quad (7)$$

with target sector $(n_\alpha \pm 1, n_\beta)$ for $\sigma_p = 0$ and $(n_\alpha, n_\beta \pm 1)$ for $\sigma_p = 1$, returning `None` when the target sector is empty ($n_\alpha = n$ for a creation, $n_\alpha = 0$ for an annihilation). These routines are adjoint pairs and satisfy the fermionic anticommutation relations across spins (verified numerically before use). The Hamiltonian $H(h_1, \text{eri})$ of a sector is applied as

$$H|v\rangle = \text{contract_2e}(\text{absorb_h1e}(h_1, \text{eri}, n, (n_\alpha, n_\beta), \tfrac{1}{2}), v), \quad H|v\rangle = 0 \text{ if } n_\alpha + n_\beta = 0, \quad (8)$$

and the dense Hamiltonian matrix of a sector is obtained by applying Eq. (8) to every unit vector and symmetrizing, $H \leftarrow \frac{1}{2}(H + H^T)$. Overlaps $\langle v|w\rangle$ are `numpy.vdot` of the flattened arrays.

2 Dyall reference (dyall.py)

2.1 Molecule and CAS calculation

- Linear H_4 : atoms at $(0, 0, kr)$, $k = 0, \dots, 3$, $r = 1.0 \text{ \AA}$ (`build_h4`); RHF followed by CASSCF($n_{\text{el}}, n_{\text{cas}}$) (`run_casscf`).
- Ozone: $R(\text{O}-\text{O}) = 1.278 \text{ \AA}$ and $\theta = 116.8^\circ$; the central O is at the origin and the terminal O atoms at $(0, \pm R \sin \frac{\theta}{2}, -R \cos \frac{\theta}{2})$; point-group symmetry is switched on (`build_o3`). RHF is followed by CASCI in the RHF orbitals with PySCF's default choice of the n_{cas} orbitals starting at $n_{\text{core}} = (N - n_{\text{el}})/2$ (`run_casci`).

2.2 Dyall Fock operator and canonicalization

With the spatial CAS one-particle density γ^{cas} (`mc.fcisolver.make_rdm1`) the spatial MO density is $\gamma^{\text{MO}} = \text{diag}(2 \cdot \mathbf{1}_{\text{core}}, \gamma^{\text{cas}}, 0)$, and

$$F^{\text{AO}} = h^{\text{AO}} + J[\gamma^{\text{AO}}] - \tfrac{1}{2}K[\gamma^{\text{AO}}], \quad \gamma^{\text{AO}} = C\gamma^{\text{MO}}C^T, \quad F^{\text{MO}} = C^T F^{\text{AO}}C, \quad (9)$$

where $J - \frac{1}{2}K$ is `mf.get_veff(mol, dm)` of the RHF object. In spin orbitals this is $F_{PQ} = h_{PQ} + \sum_{rs} \bar{v}_{Pr, Qs} \gamma_{rs}$ with the Dyall density of Eq. (18). The core-core and virtual-virtual blocks of F^{MO} are diagonalized, each irrep separately when orbital symmetry labels are available (so that the labels survive), the eigenvectors are sorted by eigenvalue, and the core and virtual columns of C are rotated accordingly; active orbitals are not changed. F^{MO} , h^{MO} and the ERIs are then recomputed in the rotated basis, and

$$\epsilon_p = F_{p_s p_s}^{\text{MO}} \quad (\text{used for } p \in \mathcal{I}). \quad (10)$$

2.3 Dyall partition

The active Dyall Hamiltonian $\hat{H}_D^{\text{act}}$ has the spatial one-body part (core index k)

$$h_{xy}^{\text{eff}} = h_{xy}^{\text{MO}} + \sum_k [2(xy|kk) - (xk|ky)] \quad (11)$$

(checked against `mc.get_h1eff`) and the all-active ERIs $(xy|zw)$. The full-space one- and two-body parts of $\hat{H}_D$, used only for exact checks, are

$$h_{p_s q_s}^D = \begin{cases} \epsilon_p \delta_{p_s q_s} & p_s, q_s \text{ inactive} \\ h_{p_s q_s}^{\text{eff}} & p_s, q_s \text{ active} \\ 0 & \text{otherwise} \end{cases}, \quad \text{eri}_{p_s r_s q_s s_s}^D = \begin{cases} (p_s r_s | q_s s_s) & \text{all four active} \\ 0 & \text{otherwise} \end{cases}. \quad (12)$$

The one-body residual interaction in spin orbitals is

$$u_{pq} = h_{pq} - h_{pq}^D \quad (h^D \text{ expanded to spin orbitals as in Eq. (2)}), \quad (13)$$

which reproduces WFL Eqs. (16)–(17) because the inactive blocks of F^{MO} are diagonal after canonicalization.

2.4 Active-space sectors and charged poles

With (n_a, n_b) the active electron numbers, the sectors

$$\begin{aligned} \mathcal{S}_0 &= (n_a, n_b), & \mathcal{S}_+ &= \{(n_a+1, n_b), (n_a, n_b+1)\} \cap \{n_\sigma \leq n_{\text{cas}}\}, \\ \mathcal{S}_- &= \{(n_a-1, n_b), (n_a, n_b-1)\} \cap \{n_\sigma \geq 0\} \end{aligned} \quad (14)$$

are diagonalized densely with $H(h^{\text{eff}}, \text{eri}^{\text{cas}})$: $H(\mathcal{S})U(\mathcal{S}) = U(\mathcal{S}) \text{diag}(E(\mathcal{S}))$. The reference is $E_0 = E_0^{(\mathcal{S}_0)}$, $|\Xi_0\rangle = U_{:,0}^{(\mathcal{S}_0)}$ (its overlap with `mc.ci` is reported). Precomputed vectors:

$$|c_x\rangle = \hat{a}_x^\dagger |\Xi_0\rangle, \quad |d_x\rangle = \hat{a}_x |\Xi_0\rangle, \quad x \in \{0, \dots, n_{\text{act}} - 1\}, \quad (15)$$

each with its sector (or `None`). The charged poles are, for every sector $\mathcal{S} \in \mathcal{S}_\pm$ and every eigenstate k with vector $|\Xi_k\rangle$,

$$\text{attachment (sign } +1): \quad \kappa_\alpha = E_k^{(\mathcal{S})} - E_0, \quad [d_\alpha]_x = \langle c_x | \Xi_k \rangle = \langle \Xi_0 | \hat{a}_x | \Xi_k \rangle, \quad (16)$$

$$\text{removal (sign } -1): \quad \kappa_\alpha = -(E_k^{(\mathcal{S})} - E_0), \quad [d_\alpha]_x = \langle \Xi_k | d_x \rangle = \langle \Xi_k | \hat{a}_x | \Xi_0 \rangle, \quad (17)$$

where $[d_\alpha]_x$ is set to zero when the sector of c_x (respectively d_x) differs from $\mathcal{S}$. Each pole α stores (sign, sector, κ_α , $|\Xi_k\rangle$, d_α); the matrix $D_{\text{act}} = [d_\alpha]$ has shape $n_{\text{act}} \times n_{\text{poles}}$. The active sum rule $D_{\text{act}} D_{\text{act}}^T = \mathbf{1}$ follows from completeness of the $N_a \pm 1$ eigenstates and is checked.

2.5 Density, static self-energy, transition tensors

$$\gamma_{xy}^{\text{act}} = \langle d_x | d_y \rangle = \langle \Xi_0 | \hat{a}_x^\dagger \hat{a}_y | \Xi_0 \rangle \text{ (sectors equal, else 0)}, \quad \gamma_{pq} = \begin{cases} \delta_{pq} & p, q \in \mathcal{C} \\ \gamma_{x_\ell y_\ell}^{\text{act}} & p, q \in \mathcal{A} \\ 0 & \text{otherwise,} \end{cases} \quad (18)$$

$$[\Sigma_{\text{stat}}]_{pq} = u_{pq} + \sum_{rs} \bar{v}_{pq,rs}^R \gamma_{rs} \quad (\text{WFL Eq. (20), } \Sigma_1^{11}), \quad (19)$$

$$C_3[\alpha, z, y, w] = \begin{cases} \langle \hat{a}_z^\dagger \hat{a}_w^\dagger \hat{a}_y \Xi_0 | \Xi_\alpha \rangle = \langle \Xi_0 | \hat{a}_y^\dagger \hat{a}_w \hat{a}_z | \Xi_\alpha \rangle & \text{attachment poles} \\ \langle \Xi_\alpha | \hat{a}_y^\dagger \hat{a}_w \hat{a}_z \Xi_0 \rangle & \text{removal poles} \end{cases} \quad (20)$$

$$\mathcal{C}_{\alpha;z,yw} = \frac{1}{2} C_3[\alpha, z, y, w] - \gamma_{yw}^{\text{act}} [d_\alpha]_z, \quad (21)$$

where the operator strings are applied right to left with Eqs. (6)–(7) and contribute only when the resulting sector equals the pole's sector. Equation (21) is the connected tensor $\mathcal{C}_{\mu;z,yw}^{D,\pm}$ of the Dyall–Hedin notes.

2.6 Neutral transition densities

For the excited states $\mu \geq 1$ of $\mathcal{S}_0$,

$$\omega_\mu = E_\mu^{(\mathcal{S}_0)} - E_0, \quad \rho_{xy}^\mu = \langle \Xi_\mu | \hat{a}_x^\dagger \hat{a}_y | \Xi_0 \rangle = \langle \Xi_\mu | \hat{a}_x^\dagger d_y \rangle \quad (\sigma_x = \sigma_y, \text{ else } 0). \quad (22)$$

2.7 Pole representation of G_D

$$\begin{aligned} T_D &= [e_P \ (P \in \mathcal{I}) \mid d_\alpha \text{ (in the active rows)}] \in \mathbb{R}^{n_{\text{so}} \times (n_I + n_{\text{poles}})}, \\ K_D &= (\epsilon_P, \kappa_\alpha), \quad G_D(z) = T_D (z - K_D)^{-1} T_D^T, \end{aligned} \quad (23)$$

with $n_I = |\mathcal{I}|$. $T_D T_D^T = \mathbf{1}$ (zeroth moment) is checked.

3 MR-RPA screening and the MR-GW bath (mrrpa.py)

3.1 Channels of the reference polarizability

Each channel c has an energy ω_c and a transition-density matrix $\rho^c \in \mathbb{R}^{n_{\text{so}} \times n_{\text{so}}}$, $\rho_{pr}^c = \langle \Phi_c | \hat{p}^\dagger \hat{r} | \Phi_0 \rangle$:

$$(\text{cv}) \ i \in \mathcal{C}, \ a \in \mathcal{V}, \ \sigma_a = \sigma_i : \quad \rho_{ai} = 1, \quad \omega = \epsilon_a - \epsilon_i; \quad (24)$$

$$(\text{ca}) \ i \in \mathcal{C}, \ \alpha \text{ attachment} : \quad \rho_{xi} = [d_\alpha]_{x_\ell} \ (x \in \mathcal{A}), \quad \omega = \kappa_\alpha - \epsilon_i; \quad (25)$$

$$(\text{av}) \ a \in \mathcal{V}, \ \alpha \text{ removal} : \quad \rho_{ax} = [d_\alpha]_{x_\ell}, \quad \omega = \epsilon_a - \kappa_\alpha; \quad (26)$$

$$(\text{aa}) \ \mu \geq 1 : \quad \rho_{xy} = \rho_{x_\ell y_\ell}^\mu, \quad \omega = \omega_\mu. \quad (27)$$

Channels with $\|\rho^c\|_F < 10^{-10}$ are discarded (this removes the spin-forbidden ca/av combinations). The (cv) channel appears once per spin-conserving pair; the ca/av channels contain all attachment/removal poles of both S_z sectors and prune themselves through the norm test.

3.2 Direct MR-RPA equations

With K channels, $\rho_f \in \mathbb{R}^{K \times n_{\text{so}}^2}$ the flattened densities and $v_m^R \in \mathbb{R}^{n_{\text{so}}^2 \times n_{\text{so}}^2}$ the residual interaction with row pair (p, r) and column pair (q, s) ,

$$[V\rho]_{pr}^c = \sum_{qs} v_{pr,qs}^R \rho_{qs}^c, \quad [V\rho^T]_{pr}^c = \sum_{qs} v_{pr,qs}^R \rho_{sq}^c, \quad (28)$$

$$B_{cc'} = \sum_{pr} \rho_{pr}^c [V\rho]_{pr}^{c'}, \quad A_{cc'} = \omega_c \delta_{cc'} + \sum_{pr} \rho_{pr}^c [V\rho^T]_{pr}^{c'} \quad (\text{WFL Eqs. (S30)–(S31)}), \quad (29)$$

followed by $A \leftarrow \frac{1}{2}(A + A^T)$, $B \leftarrow \frac{1}{2}(B + B^T)$ (the asymmetry before symmetrization is reported). The RPA problem $\begin{pmatrix} A & B \\ B & A \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \Omega \begin{pmatrix} X \\ -Y \end{pmatrix}$ is solved as

$$A - B = LL^T \text{ (Cholesky)}, \quad L^T(A + B)LZ = Z\Omega^2, \quad (30)$$

$$R \equiv X + Y = LZ\Omega^{-1/2}, \quad S \equiv X - Y = (A + B)R\Omega^{-1}, \quad R^T S = \mathbf{1},$$

with $\Omega > 0$ and $Z^T Z = \mathbf{1}$. If $A - B$ is not positive definite the general form is used instead: eigenpairs of $\begin{pmatrix} A & B \\ -B & -A \end{pmatrix}$ with $\text{Re } \Omega > 0$, $|\text{Im } \Omega| < 10^{-8} \max(1, |\Omega|)$ and positive norm $X^T X - Y^T Y > 0$ are kept, sorted, and normalized to unit norm; a warning is issued if fewer than K such solutions exist.

3.3 Screened-interaction couplings and static W

$$\bar{\rho}_{I,qs} = \sum_c \rho_{qs}^c R_{cI}, \quad M_{pr,I} = \sum_{qs} v_{pr,qs}^R \bar{\rho}_{I,qs} \quad (\text{WFL Eq. (S43)}), \quad (31)$$

$$w_{pr,qs}^c(0) = -2 \sum_I \frac{M_{pr,I} M_{qs,I}}{\Omega_I}, \quad W_{pr,qs}^0 = v_{pr,qs}^R + w_{pr,qs}^c(0) \quad (W_D(\omega = 0)), \quad (32)$$

and the mode norms $m_I = (\sum_{pr} M_{pr,I}^2)^{1/2}$. The frequency-dependent correlation part, used only in checks, is $w_{pr,qs}^c(z) = \sum_I M_{pr,I} M_{qs,I} [(z - \Omega_I)^{-1} - (z + \Omega_I)^{-1}]$.

3.4 The MR-GW pole bath

Modes with $m_I \leq 10^{-12} \max_J m_J$ are dropped (they are decoupled from the residual interaction, e.g. spin-flip modes). For the remaining modes the retarded correlation self-energy of WFL Eq. (S52), $\Sigma^c(z) = A_B (z - E_B)^{-1} A_B^T$, is built from the columns

$$\begin{aligned} \Lambda = (a, I), \ a \in \mathcal{V} : \quad & E_\Lambda = \epsilon_a + \Omega_I, \quad [A_B]_{p\Lambda} = M_{ap,I}, \\ \Lambda = (\alpha+, I) : \quad & E_\Lambda = \kappa_\alpha + \Omega_I, \quad [A_B]_{p\Lambda} = \sum_{x \in \mathcal{A}} [d_\alpha]_{x_\ell} M_{xp,I}, \\ \Lambda = (i, I), \ i \in \mathcal{C} : \quad & E_\Lambda = \epsilon_i - \Omega_I, \quad [A_B]_{p\Lambda} = M_{pi,I}, \\ \Lambda = (\alpha-, I) : \quad & E_\Lambda = \kappa_\alpha - \Omega_I, \quad [A_B]_{p\Lambda} = \sum_{x \in \mathcal{A}} [d_\alpha]_{x_\ell} M_{px,I}, \end{aligned} \quad (33)$$

and columns with $\|[A_B]_{\cdot\Lambda}\| \leq 10^{-12} \max_{\Lambda'} \|[A_B]_{\cdot\Lambda'}\|$ are dropped. The full retarded MR-GW self-energy is

$$\Sigma^{\text{MR-GW}}(z) = \Sigma_{\text{stat}} + A_B (z - E_B)^{-1} A_B^T. \quad (34)$$

4 Hermitian pole-space matrix (hermitian.py)

4.1 Top space

With T_D and K_D from Eq. (23), $n_{\text{top}} = n_I + n_{\text{poles}}$,

$$K = \text{diag}(K_D) + T_D^T \Sigma_{\text{stat}} T_D + \begin{pmatrix} 0 & K_B \\ K_B^T & 0 \end{pmatrix}, \quad K \leftarrow \frac{1}{2}(K + K^T), \quad (35)$$

$$[K_B]_{P\alpha} = \sum_{z,y,w \in \mathcal{A}} \mathcal{I}_{Pz,yw} \mathcal{C}_{\alpha;z,yw} \quad (P \in \mathcal{I}), \quad (36)$$

$$\mathcal{I}_{Pz,yw} = \begin{cases} 0 & \text{mixed='none'} \\ \bar{v}_{Pz,yw}^R & \text{mixed='bare'} \\ v_{Pz,yw}^R - W_{Pw,yz}^0 & \text{mixed='screened'} \end{cases} \quad (\mathcal{I}_D^{\text{stat}} = v_R - [W_D(0)]^x). \quad (37)$$

The pair (P, α) is “same-sector” if $(P \in \mathcal{V}$ and α is an attachment pole) or $(P \in \mathcal{C}$ and α is a removal pole). With `cross_sector=False` the inactive-active blocks of both $T_D^T \Sigma_{\text{stat}} T_D$ and K_B are multiplied by the same-sector mask; the default keeps all pairs (Objective II). `include_static=False` drops $T_D^T \Sigma_{\text{stat}} T_D$.

4.2 Full matrix, resolvent and spectral function

With the bath of Eq. (33), $\tilde{A} = T_D^T A_B \in \mathbb{R}^{n_{\text{top}} \times n_B}$,

$$\mathbb{H} = \begin{pmatrix} K & \tilde{A} \\ \tilde{A}^T & \text{diag}(E_B) \end{pmatrix}, \quad \mathbb{T} = (T_D \ 0) \in \mathbb{R}^{n_{\text{so}} \times (n_{\text{top}} + n_B)}, \quad (38)$$

applied matrix-free as $(\mathbb{H}v)_{\text{top}} = Kv_{\text{top}} + \tilde{A}v_B$, $(\mathbb{H}v)_B = \tilde{A}^T v_{\text{top}} + E_B \circ v_B$, with diagonal $(\text{diag } K, E_B)$. Without bath ($n_B = 0$) the matrix is K . Dense diagonalization $\mathbb{H}U = U \text{diag}(E)$ gives

$$Z_\lambda = \mathbb{T}U_{\cdot,\lambda}, \quad w_\lambda = \|Z_\lambda\|^2, \quad G(z) = \sum_\lambda \frac{Z_\lambda Z_\lambda^T}{z - E_\lambda}, \quad A(\omega) = \sum_\lambda Z_\lambda Z_\lambda^T \frac{\eta/\pi}{(\omega - E_\lambda)^2 + \eta^2} \quad (39)$$

(poles, spectral_function; the trace is returned unless the full matrix is requested). Without diagonalization (large baths),

$$G(z) = T_D \left[z - K - \tilde{A} (z - E_B)^{-1} \tilde{A}^T \right]^{-1} T_D^T, \quad A(\omega) = -\frac{1}{\pi} \text{Im Tr } G(\omega + i\eta) \quad (40)$$

(greens_function, spectral_function_resolvent). Spectral moments: $M_0 = T_D T_D^T$, $M_1 = T_D K T_D^T$.

4.3 Bath with the top-space Hamiltonian in the one-boson sector

With `bath_hamiltonian='top'` the bath states are $|n\rangle \otimes |1_I\rangle$ for *all* top states n and all coupled modes I ($m_I > 10^{-12} \max_J m_J$), the top-space Hamiltonian K of Eq. (35) acts inside the one-boson sector, and the couplings are the boson vertices between top states:

$$B_I = K + \text{diag}(\sigma_n) \Omega_I, \quad [g_I]_{mn} = \sum_{pr} [T_D]_{pm} M_{pr,I} [T_D]_{rn} = t_m^T M_I t_n, \quad (41)$$

$$\mathbb{H} = \begin{pmatrix} K & g_1 & g_2 & \cdots \\ g_1^T & B_1 & & \\ g_2^T & & B_2 & \\ \vdots & & & \ddots \end{pmatrix},$$

where $\sigma_n = +1$ for virtual orbitals and attachment poles and -1 for core orbitals and removal poles. The diagonal bath of Eq. (33) is the special case in which B_I is replaced by its diagonal $\kappa_n + \sigma_n \Omega_I$ (and zero-coupling states are dropped); the two coincide when K is diagonal. The bath is applied as $(Bv)_{(nI)} = \sum_{n'} K_{nn'} v_{(n'I)} + \sigma_n \Omega_I v_{(nI)}$, the diagonal is $K_{nn} + \sigma_n \Omega_I$, and the resolvent uses

$$G(z) = T_D \left[z - K - \sum_I g_I (z - B_I)^{-1} g_I^T \right]^{-1} T_D^T. \quad (42)$$

Without cross-sector couplings ($[K, \sigma] = 0$) the bath Schur complement equals $T_D^T \Sigma^{GW} [G_{\text{init}}](z) T_D$ with

$$\Sigma^{GW} [G_{\text{init}}]_{pq}(z) = \sum_{k,I} \frac{[M_I Z_k]_p [M_I Z_k]_q}{z - \lambda_k - \sigma_k \Omega_I}, \quad K = U \text{diag}(\lambda) U^T, \quad Z_k = T_D U_{:,k}, \quad \sigma_k = U_{:,k}^T \sigma U_{:,k}, \quad (43)$$

i.e. the MR-*GW* self-energy built with the poles of the Eq. (7) propagator of the preliminary section instead of the Dyall poles (`dressed_bath_self_energy`; checked).

4.4 Guess vectors for root following

`unit_vector(k)` is e_k in the $n_{\text{top}} + n_B$ space; `index_inactive(P)` and `index_active_pole( $\alpha$ )` give the top-space positions. `orbital_guess(p, sector)` is the row $\mathbb{T}_{p,:}$ (the components of orbital p on all poles); for an active p and sector '`attach`'/'`remove`' only the active poles of that sign are kept; the vector is normalized. The two H_4 principal targets use the HOMO and LUMO spatial orbitals $p_s = N/2 - 1$ and $N/2$ (spin α).

4.5 Hall-form insertion (illustration of the non-PSD Green's function)

For $z = \omega + i\eta$, with K_B from Eq. (36) (bare kernel by default),

$$\begin{aligned} [\Sigma^{12}(z)]_{Px} &= \sum_{\alpha} \frac{[K_B]_{P\alpha} [d_{\alpha}]_{x\ell}}{z - \kappa_{\alpha}} \quad (P \in \mathcal{I}, x \in \mathcal{A}), \quad \Sigma^{21} = (\Sigma^{12})^T, \\ \Sigma^{11}(z) &= \Sigma_{\text{stat}} + A_B (z - E_B)^{-1} A_B^T, \quad M(z) = (\mathbf{1} - \Sigma^{21})^{-1} G_D(z) (\mathbf{1} - \Sigma^{12})^{-1}, \\ G_{\text{Hall}}(z) &= [M(z)^{-1} - \Sigma^{11}(z)]^{-1}. \end{aligned} \quad (44)$$

5 Davidson solver with root following (`davidson.py`)

Input: `matvec` (block application of a real symmetric $\mathbb{H}$ of dimension n), the diagonal D , m guess vectors $g_1, \dots, g_m$ (normalized), tolerance τ (scripts: 10^{-7} or 10^{-8}), maximum subspace $n_{\text{max}} = \min(n, \max(12m, 40))$, preconditioner floor $\delta = 10^{-3}$.

1. $V \leftarrow$ orthonormal columns of $[g_1 \cdots g_m]$ (QR, columns with $|R_{jj}| \leq 10^{-10}$ dropped; an error is raised if fewer than m remain); $W \leftarrow \mathbb{H}V$; reference vectors $F \leftarrow [g_j]$.
2. Iterate (≤ 200 times): $H_s = \frac{1}{2}(V^T W + W^T V)$, $H_s S = S \text{diag}(\theta)$; Ritz vectors $X = VS$, $\mathbb{H}X = WS$.
3. Selection (`mode='overlap'`): for $j = 1, \dots, m$ in order, choose the not yet selected Ritz index $l_j = \arg \max_l |F_{:,j}^T X_{:,l}|$ ('`energy`': closest θ_l to the j -th target energy; '`lowest`': the first m).
4. Residuals $r_j = (\mathbb{H}X)_{:,l_j} - \theta_{l_j} X_{:,l_j}$; converged if $\|r_j\| < \tau$ for all j ; record the history.
5. For each unconverged j : $\delta_j = -r_j \oslash (D - \theta_{l_j})$, where entries of $D - \theta$ with $|D - \theta| < \delta$ are replaced by $\pm\delta$ (sign preserved); orthogonalize δ_j twice against V and against the already accepted new vectors; accept if its norm exceeds 10^{-8} of the norm before orthogonalization (and 10^{-14}) and normalize. If the preconditioned correction lies in the current subspace (this happens when the relevant block of $\mathbb{H}$ is exactly diagonal, where $-r_j \oslash (D - \theta)$ is parallel to the Ritz vector), the residual r_j itself is orthogonalized and used instead.

6. If no new vector was accepted, stop. If $\dim V + n_{\text{new}} > n_{\text{max}}$, restart with $V \leftarrow$ orthonormal columns of $[X_{:,l_1} \cdots X_{:,l_m} \delta \dots]$ and $W \leftarrow \mathbb{H}V$; otherwise append. Set $F \leftarrow [X_{:,l_j}]$ (root following) and return to step 2.

Output: θ_{l_j} , $X_{:,l_j}$, convergence flags, iteration count, residual norms. Spectral weights of the roots are $\|\mathbb{T}X_{:,l_j}\|^2$ and their overlaps with the guesses $|g_j^T X_{:,l_j}|$.

6 Full-CI reference and validation identities (`fci_reference.py`)

6.1 Exact Green's function

For the full orbital space (n_{mo} orbitals, (N_α, N_β) electrons, integrals $h_1 = h^{\text{MO}}$, $\text{eri} = (pr|qs)$, or the $\hat{H}_D$ or $\hat{H}_D + \lambda \hat{V}_R$ integrals when given) the sectors $\mathcal{S}_0, \mathcal{S}_\pm$ of Eq. (14) are diagonalized densely, $|\Psi_0\rangle$ and E_0 are taken from $\mathcal{S}_0$, and for every spin orbital p and eigenstate k of $\mathcal{S}_\pm$

$$\text{attachment: } \kappa = E_k - E_0, \quad t_p = \langle \hat{a}_p^\dagger \Psi_0 | \Psi_k \rangle; \quad \text{removal: } \kappa = -(E_k - E_0), \quad t_p = \langle \Psi_k | \hat{a}_p \Psi_0 \rangle, \quad (45)$$

giving $T \in \mathbb{R}^{n_{\text{so}} \times n_{\text{poles}}}$, K , $G(z) = T(z - K)^{-1}T^T$, $A(\omega)$ as in Eq. (39) with $Z_\lambda \rightarrow T_{:, \lambda}$, and $M_0 = TT^T$, $M_1 = TKT^T$. “Principal poles” of a sector are the poles with the largest $\|t\|^2$ above 10^{-3} .

6.2 Embedding of active-space vectors

An active-space CI vector of sector (n_a, n_b) is mapped to the full space with $(n_a + n_{\text{core}}, n_b + n_{\text{core}})$ electrons by setting the amplitude of every determinant pair to that of its active part: for each active α -string s (`cistring.make_strings(range( $n_{\text{cas}}$ ),  $n_a$ )`) the full string is $s_{\text{full}} = (2^{n_{\text{core}}} - 1) | (s \ll n_{\text{core}})$ with address `cistring.str2addr( $n_{\text{mo}}$ ,  $n_a + n_{\text{core}}$ ,  $s_{\text{full}}$ )`, and likewise for β ; virtual orbitals are empty. Because the closed core is an even operator string, all embedded matrix elements equal the active-space ones up to one common sign per pair of sectors.

6.3 Same-sector coupling check

With K from Eq. (35) for `mixed='bare'`, no bath, and $E_0^H = \langle \Psi_0 | \hat{H} | \Psi_0 \rangle$ ($|\Psi_0\rangle$ the embedded $|\Xi_0\rangle$, $\hat{H}$ the full Hamiltonian), the following identities are tested (all in the full determinant space; $|\Psi_\alpha\rangle$ the embedded active pole state, $|w_a\rangle = \hat{a}_a^\dagger |\Psi_0\rangle$ for $a \in \mathcal{V}$, $|w_i\rangle = \hat{a}_i |\Psi_0\rangle$ for $i \in \mathcal{C}$, $\sigma_\alpha = \pm 1$ the pole sign, $\sigma_P = +1$ for virtual and -1 for core):

$$\langle \Psi_\alpha | \hat{H} | \Psi_\beta \rangle - E_0^H \delta_{\alpha\beta} = \sigma_\alpha K_{\alpha\beta} \quad (\text{same sector}), \quad (46)$$

$$\langle w_Q | \hat{H} | w_P \rangle - E_0^H \delta_{PQ} = \sigma_P K_{QP} \quad (\text{same sector}), \quad (47)$$

$$\sigma_\alpha \langle w_P | \hat{H} | \Psi_\alpha \rangle = s_\alpha K_{P\alpha}, \quad s_\alpha = \pm 1 \text{ common to all } P \text{ of the sector}, \quad (48)$$

where $s_\alpha = \text{sign}(\sum_P \langle w_P | \hat{H} | \Psi_\alpha \rangle K_{P\alpha})$ absorbs the sector-pair phase of the embedding.

6.4 First-order Green's-function check

With $h_1(\lambda) = h^D + \lambda(h^{\text{MO}} - h^D)$, $\text{eri}(\lambda) = \text{eri}^D + \lambda(\text{eri} - \text{eri}^D)$, $\lambda = 10^{-3}$, and $K^{(1)} = K - \text{diag}(K_D)$ from Eq. (35) (bare kernel, no bath, all pole pairs),

$$\frac{G_{\text{FCI}}(\lambda; z) - G_{\text{FCI}}(-\lambda; z)}{2\lambda} \stackrel{?}{=} T_D R_D(z) K^{(1)} R_D(z) T_D^T, \quad R_D(z) = (z - K_D)^{-1}, \quad (49)$$

at $z \in \{-1.2, -0.6, -0.2, 0.2, 0.6, 1.2\} + 0.05i$ Ha, together with $G_{\text{FCI}}(0; z) = G_D(z)$.

7 EKT/ERPA reference (ekt_erpa.py)

The object replaces the poles, D_{act} , κ , signs, C_3 , $\mathcal{C}$ and the neutral transition densities of the Dyall reference and delegates everything else to it. With `response='exact'` the exact CAS excitations of Eq. (22) are kept in the active-active channel and only the charged states are replaced (this isolates the effect of the EKT manifold); the default `response='erpa'` uses the ERPA of Sec. 7.2.

7.1 Charged manifolds and generalized eigenproblems

For each sign and each sector $\mathcal{S} \in \mathcal{S}_{\pm}$ the manifold is the list of CI vectors

$$\text{primary: } \{\hat{a}_x|\Xi_0\rangle\} \text{ (removal), } \{\hat{a}_x^\dagger|\Xi_0\rangle\} \text{ (attachment),} \quad (50)$$

$$\text{extended (removal): } \text{primary} \cup \{\hat{a}_y^\dagger \hat{a}_w \hat{a}_z |\Xi_0\rangle\}_{w>z, y}, \quad (51)$$

$$\text{extended (attachment): } \text{primary} \cup \{\hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger |\Xi_0\rangle\}_{w>z, y},$$

keeping only vectors that land in $\mathcal{S}$ and have norm $\geq 10^{-12}$ (operators applied right to left). With $V = [v_1 \cdots v_m]$ (flattened vectors as rows) and $\hat{H} = H(h^{\text{eff}}, \text{eri}^{\text{cas}})$,

$$A_{ij} = \langle v_i | \hat{H} - E_0 | v_j \rangle, \quad S_{ij} = \langle v_i | v_j \rangle, \quad A, S \leftarrow \text{symmetrized}, \quad (52)$$

so that for the primary removal manifold $S = \gamma^{\text{act}}$ and for the primary attachment manifold $S = \mathbf{1} - \gamma^{\text{act}T}$ (EKT metrics). Canonical orthogonalization:

$$S = U \text{diag}(s) U^T, \quad X_o = U_{:,s>\tau} \text{diag}(s_{s>\tau})^{-1/2}, \quad \tau = \begin{cases} \tau_{\text{occ}} = 10^{-8} & \text{primary} \\ \tau_{\text{lin}} \max_j s_j, \quad \tau_{\text{lin}} = 10^{-10} & \text{extended,} \end{cases} \quad (53)$$

then $X_o^T A X_o C_t = C_t \text{diag}(e)$, $C = X_o C_t$ ($C^T S C = \mathbf{1}$), and for each k

$$|\Xi_k^{\text{EKT}}\rangle = \sum_j C_{jk} |v_j\rangle, \quad \kappa_k = \pm e_k, \quad [d_k]_x = \begin{cases} \langle \Xi_k^{\text{EKT}} | \hat{a}_x \Xi_0 \rangle & \text{removal} \\ \langle \hat{a}_x^\dagger \Xi_0 | \Xi_k^{\text{EKT}} \rangle & \text{attachment,} \end{cases} \quad (54)$$

(zero when the sector of $\hat{a}_x^{(\dagger)} \Xi_0$ differs from $\mathcal{S}$). The rank per sector and the number of removed directions are reported; when the number of EKT poles equals the exact one, $\max |\kappa_{\text{sorted}}^{\text{EKT}} - \kappa_{\text{sorted}}^{\text{exact}}|$ is reported. C_3 and $\mathcal{C}$ are recomputed from Eqs. (20)–(21) with the EKT states. Active moments $M_k = \sum_\alpha \kappa_\alpha^k d_\alpha d_\alpha^T$, $k = 0, \dots, 3$, are compared with the exact ones.

7.2 ERPA in the active-active channel

Natural spin orbitals per spin block: $\gamma_{\sigma\sigma}^{\text{act}} = U_\sigma \text{diag}(n_\sigma) U_\sigma^T$, assembled into U and n over the local active spin orbitals ($\tilde{p}^\dagger = \sum_x U_{x\tilde{p}} \hat{a}_x^\dagger$). Same-spin excitation vectors and their NO transforms:

$$|u_{xy}\rangle = \hat{a}_x^\dagger \hat{a}_y |\Xi_0\rangle \quad (\sigma_x = \sigma_y, \text{ sector } \mathcal{S}_0), \quad |\tilde{u}_{pq}\rangle = \sum_{x,y} U_{xp} U_{yq} |u_{xy}\rangle, \quad (55)$$

for the pair manifold $\mathcal{P} = \{(p, q) : p \neq q, \sigma_p = \sigma_q, |n_q - n_p| > \tau_{\text{occ}}\}$. The equation-of-motion matrices

$$\mathcal{M}_{(pq),(rs)} = \langle \tilde{u}_{pq} | \hat{H} - E_0 | \tilde{u}_{rs} \rangle + \langle \tilde{u}_{sr} | \hat{H} - E_0 | \tilde{u}_{qp} \rangle, \quad \mathcal{M} \leftarrow \frac{1}{2}(\mathcal{M} + \mathcal{M}^T), \quad (56)$$

$$\mathcal{S}_{(pq),(rs)} = \langle \tilde{u}_{pq} | \tilde{u}_{rs} \rangle - \langle \tilde{u}_{sr} | \tilde{u}_{qp} \rangle, \quad (57)$$

are formed. Equation (56) equals the double commutator $\langle 0 | [E_{qp}, [\hat{H}, E_{rs}]] | 0 \rangle$ and Eq. (57) the commutator $\langle 0 | [E_{qp}, E_{rs}] | 0 \rangle$, which is $\delta_{pr} \delta_{qs} (n_q - n_p)$ in the natural-orbital basis; the deviation of $\mathcal{S}$ from this diagonal form is reported. With the excitation pairs $\mathcal{E} = \{(p, q) \in \mathcal{P} : n_q > n_p\}$, their

conjugates $\bar{\mathcal{E}} = \{(q, p)\}$, $N_{(pq)} = n_q - n_p$, $A = \mathcal{M}_{\mathcal{E}\mathcal{E}}$, $B = \mathcal{M}_{\mathcal{E}\bar{\mathcal{E}}}$ (the identities $\mathcal{M}_{\bar{\mathcal{E}}\bar{\mathcal{E}}} = A$, $\mathcal{M}_{\bar{\mathcal{E}}\mathcal{E}} = B$ are checked),

$$\tilde{A} = N^{-1/2} A N^{-1/2}, \quad \tilde{B} = N^{-1/2} B N^{-1/2}, \quad (\Omega, R, S) = \text{RPA solve of Eq. (30) for } (\tilde{A}, \tilde{B}), \quad (58)$$

$$X = N^{-1/2} \frac{R+S}{2}, \quad Y = N^{-1/2} \frac{R-S}{2},$$

which is the ERPA problem $\mathcal{M}c = \omega \mathcal{S}c$ with $c^T \mathcal{S}c = 1$. Transition densities $\rho^\nu = \mathcal{S}c^\nu$ in the NO basis and their back-transformation:

$$\tilde{\rho}_{pq}^\nu = N_{(pq)} X_{(pq)}^\nu, \quad \tilde{\rho}_{qp}^\nu = -N_{(pq)} Y_{(pq)}^\nu \quad ((p, q) \in \mathcal{E}), \quad \rho_{xy}^\nu = \sum_{pq} U_{xp} U_{yq} \tilde{\rho}_{pq}^\nu = \langle \nu | \hat{a}_x^\dagger \hat{a}_y | 0 \rangle. \quad (59)$$

(Ω_ν, ρ^ν) replace (ω_μ, ρ^μ) of Eq. (22) in the (aa) channel of Eq. (27); the (ca) and (av) channels use the EKT poles automatically. Sum rules checked against the exact transition densities over all same-spin pairs:

$$\sum_\nu [\rho_{pq}^\nu \rho_{rs}^\nu - \rho_{qp}^\nu \rho_{sr}^\nu] \text{ and } \sum_\nu \Omega_\nu [\rho_{pq}^\nu \rho_{rs}^\nu + \rho_{qp}^\nu \rho_{sr}^\nu] \quad (\text{ERPA}) = (\text{exact}). \quad (60)$$

7.3 Supermatrix variants with the EKT references

Because the object exposes the same attributes as `DyallReference`, every variant of Secs. 4–4.3 is available with an EKT reference: `MRRPA(ekt)` builds the four channels of Eq. (27) with the EKT poles in (ca) and (av) and the ERPA (or exact) excitations in (aa), and `HermitianGF(ekt, MRRPA(ekt), ...)` assembles Eqs. (35)–(40) and (41)–(43) with

$$K_D = \text{diag}(\epsilon_P, \kappa_\alpha^{\text{EKT}}), \quad T_D = (\mathbf{1}_i | D_{\text{act}}^{\text{EKT}}), \quad [K_B]_{P\alpha} = \sum_{zyw} I_{Pz,yw} \mathcal{C}_{\alpha;z,yw}^{\text{EKT}}, \quad (61)$$

$\mathcal{C}^{\text{EKT}}$ from Eqs. (20)–(21) with the EKT states. Two identities are specific to the EKT references and are checked in `run.checks.py`: (i) the first moment $T_D K T_D^T$ of the bare no-bath variant is identical to the exact-reference value in all blocks, because $\sum_\alpha |\Xi_\alpha^{\text{EKT}}\rangle \langle \Xi_\alpha^{\text{EKT}}|$ is the projector on a manifold that contains $\hat{a}_x^{(\dagger)} |\Xi_0\rangle$ (and κ^{EKT} , D^{EKT} conserve the active first moment); (ii) the same-sector elements of K equal the Hamiltonian matrix elements between the embedded EKT states and $\hat{a}_P^{(\dagger)} |\Psi_0\rangle$ in the full determinant space, Eqs. (46)–(48) with the EKT states. The working equations and results are collected in Sec. IV of `notes/mr-hedin.pdf`.

8 Density-matrix realization of the EKT/ERPA reference and cumulant truncation (`wick.py`, `rdm.py`)

With `EKTERPAREference(ref, ..., realization='rdm', cumulant_drop=())` all matrices of Sec. 7 are evaluated as contractions of the active integrals with the spin-orbital reduced density matrices (RDMs) of $|\Xi_0\rangle$ instead of by operator applications to CI vectors. Both realizations agree to 10^{-12} for exact RDMs (checked); the RDM realization is the one in which cumulant approximations can be introduced.

8.1 Spin-orbital RDMs (`rdm.spin_orbital_rdm`s)

Over the local active spin orbitals, for $k = 1, \dots, 4$,

$$D_{p_1 \dots p_k, q_1 \dots q_k}^{(k)} = \langle \Xi_0 | \hat{a}_{p_1}^\dagger \dots \hat{a}_{p_k}^\dagger \hat{a}_{q_k} \dots \hat{a}_{q_1} | \Xi_0 \rangle = \langle v_{p_1 \dots p_k} | v_{q_1 \dots q_k} \rangle, \quad |v_{q_1 \dots q_k}\rangle = \hat{a}_{q_k} \dots \hat{a}_{q_1} |\Xi_0\rangle, \quad (62)$$

so that $D^{(1)} = \gamma^{\text{act}}$ and $D_{pq,rs}^{(2)} = \langle p^\dagger q^\dagger sr \rangle$. The vectors $|v\rangle$ are generated for the $\binom{n_{\text{act}}}{k}$ sorted tuples $q_1 < \dots < q_k$ with `apply_des`, grouped by (n_a, n_b) sector, their Gram matrix is formed, and

the full tensor is filled by antisymmetry, $D^{(k)}[\pi(p), \sigma(q)] = \epsilon_\pi \epsilon_\sigma D^{(k)}[p, q]$. Checks: $D^{(1)} = \gamma^{\text{act}}$; random elements against direct operator application; partial traces $\sum_r D^{(k)}_{\dots r, \dots r} = (N - k + 1) D^{(k-1)}$. `DyallReference.rdms(order)` caches the result. Memory: n_{act}^8 doubles for $D^{(4)}$ (0.8 GB for $n_{\text{act}} = 10$).

8.2 Cumulants (`rdm.wedge`, `rdm.cumulants`, `rdm.reconstruct`)

The shuffle (Grassmann) product of an (a, a) -index tensor A and a (b, b) -index tensor B is

$$(A \wedge B)_{P,Q} = \sum_{S,T} \text{sgn}(S) \text{sgn}(T) A_{P_S, Q_T} B_{P_{S'}, Q_{T'}}, \quad (63)$$

S, T running over the a -element subsets of the $a + b$ upper and lower positions (order preserved, S', T' the complements, $\text{sgn}(S) = (-1)^{\sum_i (s_i - i)}$), so that every distinct term appears once and a product of m identical factors is overcounted by $m!$. The cumulant expansion is

$$\begin{aligned} D^{(2)} &= \frac{1}{2!} \gamma^{\wedge 2} + \lambda_2, & D^{(3)} &= \frac{1}{3!} \gamma^{\wedge 3} + \gamma \wedge \lambda_2 + \lambda_3, \\ D^{(4)} &= \frac{1}{4!} \gamma^{\wedge 4} + \frac{1}{2!} \gamma^{\wedge 2} \wedge \lambda_2 + \frac{1}{2!} \lambda_2 \wedge \lambda_2 + \gamma \wedge \lambda_3 + \lambda_4, \end{aligned} \quad (64)$$

solved for λ_k in `cumulants`; `reconstruct(rdms, drop=(4,))` returns $D^{(4)}$ with $\lambda_4 = 0$ and `drop=(3, 4)` returns $D^{(3)}$ and $D^{(4)}$ with $\lambda_3 = \lambda_4 = 0$ ($D^{(1)}, D^{(2)}$ unchanged). Checks: all $\lambda_k = 0$ for a single determinant (`determinant_vector`); all elements of $\lambda_2, \lambda_3, \lambda_4$ with indices on both halves vanish for the product state of two independent two-orbital subsystems; $\sum_r \lambda_{2,pr,qr} = (\gamma^2 - \gamma)_{pq}$; $D^{(4)}(\lambda_4 = 0) + \lambda_4 = D^{(4)}$.

8.3 Normal-ordering engine (`wick.py`)

An expression is a list of terms (coefficient, Kronecker deltas, integral factors, operator string). `normal_order` moves all creators to the left with $\hat{a}_i \hat{a}_j^\dagger = \delta_{ij} - \hat{a}_j^\dagger \hat{a}_i$, eliminates summation indices (labels starting with $_$) through the deltas, sorts creators and annihilators (with the sign of the permutation; repeated labels give zero) and combines equal terms. `evaluate` turns every term into an `einsum` of the integrals, identity matrices for deltas between free indices, and $D^{(k)}$ for a string with k creators and k annihilators (creators in order, annihilators reversed); strings with unequal numbers give zero. `hamiltonian()` is $\hat{H} = \sum h_{pq} p^\dagger q + \frac{1}{2} \sum V[p, r, q, s] p^\dagger q^\dagger s r$ with the convention of Eq. (5); `to_latex` renders an expression (Sec. 11).

8.4 EKT and ERPA matrices from RDMS (`EKTERPAReference._build_ekt_rdm`, `_erpa_matrices_rdm`)

Ket operators and free labels: removal $X = \hat{a}_y$ (1h) and $X = \hat{a}_y^\dagger \hat{a}_w \hat{a}_z$ (2h1p, $w > z$); attachment $X = \hat{a}_y^\dagger$ (1p) and $X = \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger$ (2p1h, $w > z$); the bra of each type is the Hermitian conjugate with labels $(y, w, z) \rightarrow (a, b, c)$. For every pair of types the arrays

$$S[\text{bra}, \text{ket}] = \langle X_i^\dagger X_j \rangle, \quad A[\text{bra}, \text{ket}] = \langle X_i^\dagger [\hat{H}_A, X_j] \rangle, \quad (65)$$

are evaluated over all free indices and gathered into the sector matrices of Eq. (52) with the manifold order of Sec. 7 (entries with $\langle X^\dagger X \rangle < 10^{-24}$ removed; the sector of an entry follows from the spins of its indices). The commutators are

$$[\hat{H}_A, \hat{a}_y] = - \sum_q h_{yq} \hat{a}_q - \sum_{qrs} v_{yr,qs} \hat{a}_q^\dagger \hat{a}_s \hat{a}_r, \quad [\hat{H}_A, \hat{a}_y^\dagger] = \sum_p h_{py} \hat{a}_p^\dagger + \sum_{pqs} v_{py,qs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s, \quad (66)$$

$$\begin{aligned} [\hat{H}_A, \hat{a}_y^\dagger \hat{a}_w \hat{a}_z] &= \sum_p h_{py} \hat{a}_p^\dagger \hat{a}_w \hat{a}_z - \sum_q h_{wq} \hat{a}_y^\dagger \hat{a}_q \hat{a}_z + \sum_q h_{zq} \hat{a}_y^\dagger \hat{a}_q \hat{a}_w - \sum_{rs} v_{zr,ws} \hat{a}_y^\dagger \hat{a}_s \hat{a}_r \\ &+ \sum_{pqs} v_{py,qs} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_w \hat{a}_z - \sum_{qrs} v_{wr,qs} \hat{a}_y^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r \hat{a}_z + \sum_{qrs} v_{zr,qs} \hat{a}_y^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r \hat{a}_w, \end{aligned} \quad (67)$$

and the 2p1h one follows by Hermitian conjugation. Resulting RDM orders: 1h–1h $D^{(1)}, D^{(2)}$ (3 terms); 1h–2h1p $D^{(2)}, D^{(3)}$ (11 terms); 2h1p–1h (5); 2h1p–2h1p $D^{(2)}-D^{(4)}$ (28); metrics $S_{x,(yzw)} = D_{xy,zw}^{(2)}$ and $S_{(abc),(yzw)} = D_{bcy,zwa}^{(3)} - \delta_{ay} D_{bc,zw}^{(2)}$. Residues: $d_{x,\alpha} = \sum_j c_j^\alpha \langle X_j^\dagger \hat{a}_x \rangle$ (removal) or $\sum_j c_j^\alpha \langle \hat{a}_x X_j \rangle$ (attachment); three-operator amplitudes $C_3[\alpha, z, y, w] = \sum_j c_j^\alpha \langle X_j^\dagger \hat{a}_y^\dagger \hat{a}_w \hat{a}_z \rangle$ (removal) or $\sum_j c_j^\alpha \langle \hat{a}_y^\dagger \hat{a}_w \hat{a}_z X_j \rangle$ (attachment), with $D^{(2)}$ (1h/1p rows) and $D^{(3)}$ (2h1p/2p1h rows); $\mathcal{C}$ from Eq. (21). The irrep of a pole is that of $|\Xi_0\rangle$ times the irreps of the orbitals of its dominant manifold operator. ERPA: $h, V, D^{(1)}, D^{(2)}$ are transformed to the natural-orbital basis and $\mathcal{M}_{(pq),(rs)} = \langle [\hat{E}_{qp}, [\hat{H}_A, \hat{E}_{rs}]] \rangle$ (20 terms, $D^{(1)}, D^{(2)}$) and $\mathcal{S}_{(pq),(rs)} = \langle [\hat{E}_{qp}, \hat{E}_{rs}] \rangle$ replace Eqs. (56)–(57); the rest of Sec. 7.2 is unchanged. All term lists are given in Sec. 11.

8.5 Cumulant truncation (cumulant_comparison.py)

`cumulant_drop=(4,)` replaces $D^{(4)}$ by its reconstruction with $\lambda_4 = 0$ (only the 2h1p–2h1p and 2p1h–2p1h Hamiltonian blocks change); `cumulant_drop=(3, 4)` also replaces $D^{(3)}$ ($\lambda_3 = 0$: the 2h1p metric, the 1h–2h1p blocks and the three-operator amplitudes change as well). The reconstructed RDMs are not N -representable: the Gram matrix may have negative eigenvalues (reported in `metric_negative`), which the canonical orthogonalization (53) discards together with the directions below $\tau_{\text{lin}} \max_j s_j$; poles may appear on the wrong side of the Fermi level, and the MR-RPA built from such poles may be unstable, in which case `solve_rpa` falls back to the non-Hermitian solver and `MRRPA.nmodes_lost` counts the complex or negative-norm modes. The script reports $\|D^{(k)}\|$, $\|\lambda_k\|$, the numbers of poles, dropped directions, negative metric eigenvalues, wrong-sign poles and lost RPA modes, the deviation of the sorted poles and of M_0-M_3 from the exact ones, and the supermatrix variants (Eq. (7) without bath, MR-GW, diagonal bath screened, dressed bath bare and screened) for the realizations CI, RDM exact, $\lambda_4 = 0$, $\lambda_3 = \lambda_4 = 0$ with $\tau_{\text{lin}} = 10^{-10}$ and with $\tau_{\text{lin}} = 5 \times 10^{-2}$; variants of an unstable reference (lost MR-RPA modes) are skipped. Poles on the wrong side of the Fermi level are reported but do not by themselves prevent the construction: for the CASCI reference of $\text{O}_3(6,4)$ in RHF orbitals the active Dyll Hamiltonian has two $N_a + 1$ states below the neutral ground state already with exact RDMs.

9 Workflows of the scripts

9.1 run_checks.py (H_4 , STO-6G, CAS(2,2), CASSCF)

The 77 checks, with the identity tested and the tolerance:

identity	tolerance
Eq. (11) vs <code>mc.get_h1eff</code> ; $E_0 + E_{\text{core}} = E_{\text{CASSCF}}$; $ \langle \Xi_0 \text{mc.ci} \rangle = 1$	$10^{-10}, 10^{-8}, 10^{-8}$
$D_{\text{act}} D_{\text{act}}^T = \mathbf{1}$; $T_D T_D^T = \mathbf{1}$	10^{-10}
Σ_{stat} : active-active, core-core, virtual-virtual blocks zero; Hermitian	$10^{-10}, 10^{-8}, 10^{-8}, 10^{-10}$
MR-RPA: A, B symmetric; $R^T S = \mathbf{1}$; $\Omega > 0$; $W_{pr,qs}^0 = W_{qs,pr}^0$	$10^{-10}, 10^{-8}, 0, 10^{-10}$
Eq. (40) with $K_B = 0 = [G_D^{-1} - \Sigma^{\text{MR-GW}}]^{-1}$ at four z	10^{-9}
Eqs. (46)–(48)	10^{-9}
$G_{\text{FCI}}(\hat{H}_D) = G_D$; Eq. (49) (relative)	$10^{-8}, 10^{-5}$
screened mixed variant: $M_0 = \mathbf{1}$; $w_\lambda \geq 0$; M_1 (bare mixed) $= h + \bar{v}\gamma_D$	$10^{-10}, 10^{-12}, 10^{-9}$
Davidson roots = dense max-overlap eigenvalues; residuals	10^{-7}
dressed bath: equals the diagonal bath for diagonal K ; equals	$10^{-10}, 10^{-9}, 10^{-10}, 10^{-12}, 10^{-7}$
Eq. (43) without cross-sector couplings; $M_0 = \mathbf{1}$; $w_\lambda \geq 0$; Davidson vs dense	
EKT (primary): $M_0 = \mathbf{1}$, M_1 exact, poles exact (CAS(2,2))	$10^{-10}, 10^{-9}, 10^{-9}$
EKT (extended): $M_0 \dots M_3$ exact; poles exact	10^{-8}
ERPA sum rules, Eq. (60)	10^{-9}
EKT/ERPA: linearization identity; $M_0 = \mathbf{1}$; $w_\lambda \geq 0$; Davidson vs dense	$10^{-9}, 10^{-10}, 10^{-12}, 10^{-7}$
EKT (primary, extended): M_1 of the bare no-bath variant = exact-reference value; same-sector K vs embedded EKT states (active-active, inactive-active)	10^{-9}
EKT (primary, extended): dressed bath = Eq. (43) without cross-sector couplings; dressed bare, dressed screened, diagonal bare: $M_0 = \mathbf{1}$ and $w_\lambda \geq 0$; Davidson vs dense	$10^{-9}, 10^{-10}, 10^{-7}$
RDMs: $D^{(1)} = \gamma$; partial traces; λ_2 trace relation; cumulants vanish for a determinant; cross-subsystem cumulants vanish for a product state; $D^{(4)}(\lambda_4 = 0) + \lambda_4 = D^{(4)}$	$10^{-12}, 10^{-12}, 10^{-12}, 10^{-14}, 10^{-14},$ 10^{-12}
RDM realization = CI realization (primary, extended): A, S , eigenvalues; ERPA energies; $G(z)$ of the dressed screened variant	$10^{-9}, 10^{-10}, 10^{-9}$
extended EKT with $\lambda_4 = 0$: M_0 exact	10^{-10}

9.2 example_h4.py

1. Build H_4 (default STO-6G, $r = 1.0 \text{ \AA}$, CAS(2,2), CASSCF), the Dyal reference, the MR-RPA, and the EKT/ERPA references with the primary and the extended charged manifold, each with its own MR-RPA.
2. Build eight Hermitian matrices: CAS (`mixed='none'`, no bath), MR-GW (`'none'`, bath), mixed bare, mixed screened, and the EKT/ERPA and EKT(2h1p)/ERPA versions of MR-GW and of the screened mixed variant; the full-CI reference (unless `--no-fci`).
3. For every matrix, Davidson root following from the guesses `orbital_guess(HOMO, 'remove')`, `orbital_guess(LUMO, 'attach')`, the unit vector of the first core spin orbital (if any) and of the first virtual spin orbital (if any); compare with the dense eigenvalue of maximal overlap with the same guess; report weights and overlaps.
4. Print the active-space excitation energies (exact and ERPA), the charged poles (exact, primary EKT, extended EKT), the manifold ranks, the four most intense poles per sector of every variant and of full CI, the sum rules M_0 , minimal weight, and $\max |M_1 - M_1^{\text{FCI}}|$ over the full matrix and over the inactive-active block.
5. Plot $\text{Tr } A(\omega)$ for $\omega \in [-30, 25] \text{ eV}$ with $\eta = 0.1 \text{ eV}$ (Eq. (39)), one panel per variant plus the Hall-form insertion of Eq. (44), full CI dashed, to `h4-<basis>_cas<n>_<m>_spectral_functions.png`.

9.3 ginit_prelim.py

1. H₄ (STO-6G CAS(2,2), 6-31G CAS(2,2), 6-31G CAS(4,4); CASSCF) or, with `--o3`, ozone (6-31G, CASCI(6,4) and (8,5) in RHF orbitals; `--cas` selects).
2. For each reference selected by `--reference` (`cas`: exact Dyall states; `ekt`: primary EKT manifold; `ekt2`: extended manifold; default all three; `--response exact` keeps the exact CAS excitations with the EKT poles) build the MR-RPA and ten Hermitian matrices: reference poles with Σ_{stat} (no bath); Eq. (7) of the notes with same-sector couplings only and with all pairs (bare kernel, no bath); MR-GW ($K_B = 0$, diagonal bath); diagonal bath with bare and screened K_B ; dressed bath with bare and screened K_B ; dressed bath without cross-sector couplings, bare and screened.
3. Davidson root following from the HOMO-like removal and LUMO-like attachment guesses plus the first core and virtual spin orbitals (H₄), or from the lowest cation pole of each irrep in the sector $(n_a - 1, n_b)$ of the respective reference (ozone); print energies, weights, matrix dimension, $|M_0 - \mathbf{1}|$, and for matrices with $n \leq 6000$ the minimal spectral weight and the deviation from the dense max-overlap eigenvalue; full CI (H₄) or the MR-GW, DMRG and experimental values of WFL (ozone) for comparison. These runs produce Tables I and II of `notes/mr-hedin.pdf`; the outputs are kept in `results/`.

9.4 cumulant_comparison.py

H₄ (6-31G, CAS(4,4)) or, with `--o3`, ozone (6-31G, CASCI(6,4) and (8,5)): RDMs $D^{(1)}-D^{(4)}$ and cumulant norms; the five realizations of Sec. 8 of the extended EKT/ERPA reference with their MR-RPA; agreement of the primary EKT and of the ERPA between the CI and RDM realizations; the pole/metric/RPA diagnostics table; the supermatrix variants by Davidson root following (H₄: HOMO-like removal and LUMO-like attachment guesses; ozone: the lowest cation pole of each irrep, whose irrep is assigned from the dominant manifold operator); full CI or DMRG for comparison. Outputs in `results/cumulant_comparison*.txt`.

9.5 check_ozone.py

1. Build O₃ (6-31G), RHF; print the orbital energies of orbitals 6–14 with C_{2v} labels (running index per irrep) against WFL Table S3; CASCI with $(n_{\text{el}}, n_{\text{cas}})$ from `--cas` (default 6,4; 8,5 supported; 8,6 refused unless `--force` because the estimated bath exceeds 3×10^6 poles); print the three leading CI coefficients.
2. Dyall reference, MR-RPA; EKT/ERPA references (primary, extended) and their MR-RPAs.
3. Cation states: for each reference, in the sector $(n_a - 1, n_b)$, the lowest removal pole of each irrep, the irrep of a pole being the XOR of the orbital irreps (PySCF ids) occupied in its dominant determinant.
4. For the variants MR-GW, mixed bare, mixed screened (exact reference), EKT/ERPA MR-GW and its screened mixed version, and the two EKT(2h1p)/ERPA versions: Davidson root following from the unit vectors of the 2A_1 , 2B_2 , 2A_2 cation poles of the respective reference; IPs = $-\theta$, weights, overlaps.
5. Table of IPs against WFL Table S5 (KT, GW@RHF, CAS, MR-GW, EOM-CCSD, DMRG, experiment) with the state ordering, and the deviation MR-GW(this work) – MR-GW(WFL).
6. Optional (`--plot`): Tr $A(\omega)$ on $[-20, -8]$ eV with $\eta = 0.1$ eV from Eq. (40) for the CASCI reference and the three exact-reference variants, with the experimental (dotted) and DMRG (dashed) IPs as vertical lines.

10 Equation-to-function index

equations	function / method	module
(1)–(5)	<code>spin_orbital_one_body</code> , <code>spin_orbital_two_body</code> , <code>antisymmetrize</code> , <code>DyallReference.__init__</code>	<code>dyall.py</code>
(6)–(8)	<code>apply_cre</code> , <code>apply_des</code> , <code>apply_hamiltonian</code> , <code>dense_ci_hamiltonian</code>	<code>dyall.py</code>
(9)–(10)	<code>DyallReference.__init__</code> , <code>_canonicalize_block</code>	<code>dyall.py</code>
(11)–(13)	<code>DyallReference.__init__</code>	<code>dyall.py</code>
(14)–(17)	<code>DyallReference.__init__</code> , <code>Sector</code> , <code>ActivePole</code>	<code>dyall.py</code>
(18)–(21)	<code>DyallReference.__init__</code> , <code>_three_operator_amplitudes</code>	<code>dyall.py</code>
(22)	<code>neutral_transition_densities</code>	<code>dyall.py</code>
(23)	<code>pole_representation</code> , <code>dyall_greens_function</code>	<code>dyall.py</code>
(27)	<code>MRRPA._build_channels</code>	<code>mrrpa.py</code>
(29)–(30)	<code>MRRPA._solve</code> , <code>solve_rpa</code> , <code>_solve_rpa_general</code>	<code>mrrpa.py</code>
(31)–(32)	<code>MRRPA._solve</code>	<code>mrrpa.py</code>
(33)–(34)	<code>MRRPA.bath</code> , <code>MRRPA.self_energy</code>	<code>mrrpa.py</code>
(35)–(37)	<code>HermitianGF.__init__</code> , <code>mixed_kernel</code>	<code>hermitian.py</code>
(38)–(40)	<code>matvec</code> , <code>diagonal</code> , <code>dense</code> , <code>eig</code> , <code>poles</code> , <code>greens_function</code> , <code>spectral_function(_resolvent)</code> , <code>moments</code> , <code>dense_roots</code>	<code>hermitian.py</code>
(41)–(43)	<code>HermitianGF.__init__</code> (<code>bath_hamiltonian='top'</code>), <code>_bath.apply</code> , <code>_bath_block</code> , <code>dressed_bath_self_energy</code>	<code>hermitian.py</code>
(44)	<code>hall_insertion_gf</code>	<code>hermitian.py</code>
Sec. 5	<code>davidson_root_following</code>	<code>davidson.py</code>
(45)	<code>FCIReference</code>	<code>fci_reference.py</code>
embedding, (46)–(49)	<code>embed_active_vector</code> , <code>check_same_sector_couplings</code> , <code>check_first_order_gf</code>	<code>fci_reference.py</code>
(50)–(54)	<code>EKTERPARReference._build_ekt</code>	<code>ekt_erpa.py</code>
(55)–(60)	<code>EKTERPARReference._build_erpa</code> , <code>neutral_transition_densities</code> , <code>response_sum_rules</code> , <code>active_moments</code>	<code>ekt_erpa.py</code>
(61)	<code>HermitianGF</code> with an <code>EKTERPARReference</code> ; <code>driver_ginit_prelim.py</code>	<code>hermitian.py</code>
(62)	<code>spin_orbital_rdms</code> , <code>DyallReference.rdms</code>	<code>rdm.py</code>
(63)–(64)	<code>wedge</code> , <code>cumulants</code> , <code>reconstruct</code> , <code>cumulant_norms</code>	<code>rdm.py</code>
(65)–(67), Sec. 11	<code>normal_order</code> , <code>evaluate</code> , <code>to_latex</code> ; <code>EKTERPARReference._build_ekt_rdm</code> , <code>_erpa_matrices_rdm</code> , <code>pole_symmetry</code>	<code>wick.py</code> , <code>ekt_erpa.py</code>

11 Explicit RDM contractions of all EKT and ERPA blocks

Generated by `wick.to_latex`; $v_{pr,qs} = (pr|qs)$, primed indices are summed over the active spin orbitals, $D^{(k)}$ as in Eq. (62), and δ between two free indices is a Kronecker delta. The bra operators are $\hat{a}_x^\dagger / \hat{a}_x$ and the Hermitian conjugates of $\hat{a}_a^\dagger \hat{a}_b \hat{a}_c / \hat{a}_a \hat{a}_b^\dagger \hat{a}_c^\dagger$.

Removal, metric $S_{x,y} = \langle \hat{a}_x^\dagger \hat{a}_y \rangle$ (1 terms, RDM orders [1]):

$$+ D_{x,y}^{(1)}$$

Removal, metric $S_{x,(y wz)} = \langle \hat{a}_x^\dagger \hat{a}_y^\dagger \hat{a}_w \hat{a}_z \rangle$ (1 terms, RDM orders [2]):

$$+ D_{xy,zw}^{(2)}$$

Removal, metric $S_{(abc),(y wz)} = \langle \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_a \hat{a}_y^\dagger \hat{a}_w \hat{a}_z \rangle$ (2 terms, RDM orders [2, 3]):

$$- \delta_{ay} D_{bc,zw}^{(2)} + D_{bcy,zwa}^{(3)}$$

Removal, $A_{x,y} = \langle \hat{a}_x^\dagger [\hat{H}_A, \hat{a}_y] \rangle$ (3 terms, RDM orders [1, 2]):

$$- h_{yq'} D_{x,q'}^{(1)} - \frac{1}{2} v_{yr',q's'} D_{q'x,s'r'}^{(2)} + \frac{1}{2} v_{p'r',ys'} D_{p'x,s'r'}^{(2)}$$

Removal, $A_{x,(y wz)} = \langle \hat{a}_x^\dagger [\hat{H}_A, \hat{a}_y^\dagger \hat{a}_w \hat{a}_z] \rangle$ (11 terms, RDM orders [2, 3]):

$$\begin{aligned} & - h_{p'y} D_{p'x,zw}^{(2)} + \frac{1}{2} v_{p'y,q's'} D_{p'q'x,zws'}^{(3)} - \frac{1}{2} v_{p'r',q'y} D_{p'q'x,zwr'}^{(3)} \\ & + h_{zq'} D_{xy,wq'}^{(2)} - h_{wq'} D_{xy,zq'}^{(2)} + \frac{1}{2} v_{zr',ws'} D_{xy,s'r'}^{(2)} \\ & - \frac{1}{2} v_{zr',q's'} D_{q'xy,ws'r'}^{(3)} - \frac{1}{2} v_{wr',zs'} D_{xy,s'r'}^{(2)} + \frac{1}{2} v_{wr',q's'} D_{q'xy,zs'r'}^{(3)} \\ & + \frac{1}{2} v_{p'r',zs'} D_{p'xy,ws'r'}^{(3)} - \frac{1}{2} v_{p'r',ws'} D_{p'xy,zs'r'}^{(3)} \end{aligned}$$

Removal, $A_{(abc),y} = \langle \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_a [\hat{H}_A, \hat{a}_y] \rangle$ (5 terms, RDM orders [2, 3]):

$$\begin{aligned} & - h_{yq'} D_{bc,aq'}^{(2)} - \frac{1}{2} v_{yr',as'} D_{bc,s'r'}^{(2)} + \frac{1}{2} v_{yr',q's'} D_{q'bc,as'r'}^{(3)} \\ & + \frac{1}{2} v_{ar',ys'} D_{bc,s'r'}^{(2)} - \frac{1}{2} v_{p'r',ys'} D_{p'bc,as'r'}^{(3)} \end{aligned}$$

Removal, $A_{(abc),(y wz)} = \langle \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_a [\hat{H}_A, \hat{a}_y^\dagger \hat{a}_w \hat{a}_z] \rangle$ (28 terms, RDM orders [2, 3, 4]):

$$\begin{aligned} & - h_{ay} D_{bc,zw}^{(2)} + h_{p'y} D_{p'bc,zwa}^{(3)} - \frac{1}{2} v_{ay,q's'} D_{q'bc,zws'}^{(3)} \\ & + \frac{1}{2} v_{ar',q'y} D_{q'bc,zwr'}^{(3)} + \frac{1}{2} v_{p'y,as'} D_{p'bc,zws'}^{(3)} - \frac{1}{2} v_{p'r',ay} D_{p'bc,zwr'}^{(3)} \\ & + \frac{1}{2} v_{p'y,q's'} D_{p'q'bc,zwas'}^{(4)} - \frac{1}{2} v_{p'r',q'y} D_{p'q'bc,zwar'}^{(4)} \\ & - \delta_{ay} h_{zq'} D_{bc,wq'}^{(2)} + \delta_{ay} h_{wq'} D_{bc,zq'}^{(2)} - h_{zq'} D_{bcy,waq'}^{(3)} \\ & + h_{wq'} D_{bcy,zaq'}^{(3)} - \frac{1}{2} \delta_{ay} v_{zr',ws'} D_{bc,s'r'}^{(2)} \\ & + \frac{1}{2} \delta_{ay} v_{zr',q's'} D_{q'bc,ws'r'}^{(3)} + \frac{1}{2} \delta_{ay} v_{wr',zs'} D_{bc,s'r'}^{(2)} \\ & - \frac{1}{2} \delta_{ay} v_{wr',q's'} D_{q'bc,zs'r'}^{(3)} - \frac{1}{2} \delta_{ay} v_{p'r',zs'} D_{p'bc,ws'r'}^{(3)} \\ & + \frac{1}{2} \delta_{ay} v_{p'r',ws'} D_{p'bc,zs'r'}^{(3)} + \frac{1}{2} v_{zr',ws'} D_{bcy,as'r'}^{(3)} \\ & - \frac{1}{2} v_{zr',as'} D_{bcy,ws'r'}^{(3)} - \frac{1}{2} v_{zr',q's'} D_{q'bcy,was'r'}^{(4)} \\ & - \frac{1}{2} v_{wr',zs'} D_{bcy,as'r'}^{(3)} + \frac{1}{2} v_{wr',as'} D_{bcy,zs'r'}^{(3)} \\ & + \frac{1}{2} v_{wr',q's'} D_{q'bcy,zas'r'}^{(4)} + \frac{1}{2} v_{ar',zs'} D_{bcy,ws'r'}^{(3)} \\ & - \frac{1}{2} v_{ar',ws'} D_{bcy,zs'r'}^{(3)} + \frac{1}{2} v_{p'r',zs'} D_{p'bcy,was'r'}^{(4)} \\ & - \frac{1}{2} v_{p'r',ws'} D_{p'bcy,zas'r'}^{(4)} \end{aligned}$$

Attachment, metric $S_{x,y} = \langle \hat{a}_x \hat{a}_y^\dagger \rangle$ (2 terms, RDM orders [0, 1]):

$$+ \delta_{xy} - D_{y,x}^{(1)}$$

Attachment, metric $S_{x,(y wz)} = \langle \hat{a}_x \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger \rangle$ (7 terms, RDM orders [0, 1, 2]):

$$+ \delta_{wy} \delta_{xz} - \delta_{wy} D_{z,x}^{(1)} - \delta_{wx} \delta_{yz} + \delta_{wx} D_{z,y}^{(1)}$$

$$+ \delta_{yz} D_{w,x}^{(1)} - \delta_{xz} D_{w,y}^{(1)} + D_{wz,yx}^{(2)}$$

Attachment, metric $S_{(abc),(y wz)} = \langle \hat{a}_c \hat{a}_b \hat{a}_a^\dagger \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger \rangle$ (27 terms, RDM orders [0, 1, 2, 3]):

$$\begin{aligned} & + \delta_{ab} \delta_{cz} \delta_{wy} - \delta_{ab} \delta_{wy} D_{z,c}^{(1)} - \delta_{ab} \delta_{cw} \delta_{yz} \\ & + \delta_{ab} \delta_{cw} D_{z,y}^{(1)} + \delta_{ab} \delta_{yz} D_{w,c}^{(1)} - \delta_{ab} \delta_{cz} D_{w,y}^{(1)} \\ & + \delta_{ab} D_{wz,yc}^{(2)} - \delta_{ac} \delta_{bz} \delta_{wy} + \delta_{ac} \delta_{wy} D_{z,b}^{(1)} \\ & + \delta_{ac} \delta_{bw} \delta_{yz} - \delta_{ac} \delta_{bw} D_{z,y}^{(1)} - \delta_{ac} \delta_{yz} D_{w,b}^{(1)} \\ & + \delta_{ac} \delta_{bz} D_{w,y}^{(1)} - \delta_{ac} D_{wz,yb}^{(2)} + \delta_{bz} \delta_{wy} D_{a,c}^{(1)} \\ & - \delta_{cz} \delta_{wy} D_{a,b}^{(1)} - \delta_{wy} D_{az,cb}^{(2)} - \delta_{bw} \delta_{yz} D_{a,c}^{(1)} \\ & + \delta_{bw} \delta_{cz} D_{a,y}^{(1)} - \delta_{bw} D_{az,yc}^{(2)} + \delta_{cw} \delta_{yz} D_{a,b}^{(1)} \\ & - \delta_{bz} \delta_{cw} D_{a,y}^{(1)} + \delta_{cw} D_{az,yb}^{(2)} + \delta_{yz} D_{aw,cb}^{(2)} + \delta_{bz} D_{aw,yc}^{(2)} \\ & - \delta_{cz} D_{aw,yb}^{(2)} - D_{awz,ycb}^{(3)} \end{aligned}$$

Attachment, $A_{x,y} = \langle \hat{a}_x [\hat{H}_A, \hat{a}_y^\dagger] \rangle$ (8 terms, RDM orders [0, 1, 2]):

$$\begin{aligned} & + h_{xy} - h_{p'y} D_{p',x}^{(1)} + \frac{1}{2} v_{xy,q's'} D_{q',s'}^{(1)} - \frac{1}{2} v_{xr',q'y} D_{q',r'}^{(1)} \\ & - \frac{1}{2} v_{p'y,xs'} D_{p',s'}^{(1)} + \frac{1}{2} v_{p'r',xy} D_{p',r'}^{(1)} - \frac{1}{2} v_{p'y,q's'} D_{p'q',xs'}^{(2)} \\ & + \frac{1}{2} v_{p'r',q'y} D_{p'q',xr'}^{(2)} \end{aligned}$$

Attachment, $A_{x,(y wz)} = \langle \hat{a}_x [\hat{H}_A, \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger] \rangle$ (55 terms, RDM orders [0, 1, 2, 3]):

$$\begin{aligned} & + \delta_{wy} h_{xz} - \delta_{yz} h_{xw} + h_{xw} D_{z,y}^{(1)} - h_{xz} D_{w,y}^{(1)} - \delta_{wy} h_{p'z} D_{p',x}^{(1)} \\ & + \delta_{yz} h_{p'w} D_{p',x}^{(1)} - \delta_{xz} h_{p'w} D_{p',y}^{(1)} + h_{p'w} D_{p'z,yx}^{(2)} \\ & + \delta_{wx} h_{p'z} D_{p',y}^{(1)} - h_{p'z} D_{p'w,yx}^{(2)} + \frac{1}{2} \delta_{wy} v_{xz,q's'} D_{q',s'}^{(1)} \\ & - \frac{1}{2} \delta_{wy} v_{xr',q'z} D_{q',r'}^{(1)} - \frac{1}{2} \delta_{yz} v_{xw,q's'} D_{q',s'}^{(1)} \\ & + \frac{1}{2} v_{xw,q'z} D_{q',y}^{(1)} - \frac{1}{2} v_{xw,q's'} D_{q'z,ys'}^{(2)} + \frac{1}{2} \delta_{yz} v_{xr',q'w} D_{q',r'}^{(1)} \\ & - \frac{1}{2} v_{xz,q'w} D_{q',y}^{(1)} + \frac{1}{2} v_{xr',q'w} D_{q'z,yr'}^{(2)} + \frac{1}{2} v_{xz,q's'} D_{q'w,ys'}^{(2)} \\ & - \frac{1}{2} v_{xr',q'z} D_{q'w,yr'}^{(2)} - \frac{1}{2} \delta_{wy} v_{p'z,xs'} D_{p',s'}^{(1)} \\ & + \frac{1}{2} \delta_{wy} v_{p'r',xz} D_{p',r'}^{(1)} + \frac{1}{2} \delta_{yz} v_{p'w,xs'} D_{p',s'}^{(1)} \\ & - \frac{1}{2} v_{p'w,xz} D_{p',y}^{(1)} + \frac{1}{2} v_{p'w,xs'} D_{p'z,ys'}^{(2)} - \frac{1}{2} \delta_{yz} v_{p'r',xw} D_{p',r'}^{(1)} \\ & + \frac{1}{2} v_{p'z,xw} D_{p',y}^{(1)} - \frac{1}{2} v_{p'r',xw} D_{p'z,yr'}^{(2)} - \frac{1}{2} v_{p'z,xs'} D_{p'w,ys'}^{(2)} \\ & + \frac{1}{2} v_{p'r',xz} D_{p'w,yr'}^{(2)} - \frac{1}{2} \delta_{wy} v_{p'z,q's'} D_{p'q',xs'}^{(2)} \\ & + \frac{1}{2} \delta_{wy} v_{p'r',q'z} D_{p'q',xr'}^{(2)} + \frac{1}{2} \delta_{yz} v_{p'w,q's'} D_{p'q',xs'}^{(2)} \\ & + \frac{1}{2} v_{p'w,q'z} D_{p'q',yx}^{(2)} - \frac{1}{2} \delta_{xz} v_{p'w,q's'} D_{p'q',ys'}^{(2)} \\ & - \frac{1}{2} v_{p'w,q's'} D_{p'q'z,yxs'}^{(3)} - \frac{1}{2} \delta_{yz} v_{p'r',q'w} D_{p'q',xr'}^{(2)} \\ & - \frac{1}{2} v_{p'z,q'w} D_{p'q',yx}^{(2)} + \frac{1}{2} \delta_{xz} v_{p'r',q'w} D_{p'q',yr'}^{(2)} \\ & + \frac{1}{2} v_{p'r',q'w} D_{p'q'z,yxr'}^{(3)} + \frac{1}{2} \delta_{wx} v_{p'z,q's'} D_{p'q',ys'}^{(2)} \\ & - \frac{1}{2} \delta_{wx} v_{p'r',q'z} D_{p'q',yr'}^{(2)} + \frac{1}{2} v_{p'z,q's'} D_{p'q'w,yxs'}^{(3)} \\ & - \frac{1}{2} v_{p'r',q'z} D_{p'q'w,yxr'}^{(3)} - \delta_{wx} h_{yq'} D_{z,q'}^{(1)} + \delta_{xz} h_{yq'} D_{w,q'}^{(1)} \\ & + h_{yq'} D_{wz,xq'}^{(2)} - \frac{1}{2} \delta_{wx} v_{yr',q's'} D_{q'z,s'r'}^{(2)} \\ & + \frac{1}{2} \delta_{wx} v_{p'r',ys'} D_{p'z,s'r'}^{(2)} + \frac{1}{2} \delta_{xz} v_{yr',q's'} D_{q'w,s'r'}^{(2)} \\ & - \frac{1}{2} \delta_{xz} v_{p'r',ys'} D_{p'w,s'r'}^{(2)} + \frac{1}{2} v_{yr',xs'} D_{wz,s'r'}^{(2)} \end{aligned}$$

$$- \frac{1}{2} v_{yr',q's'} D_{q'wz,xs'r'}^{(3)} - \frac{1}{2} v_{xr',ys'} D_{wz,s'r'}^{(2)} + \frac{1}{2} v_{p'r',ys'} D_{p'wz,xs'r'}^{(3)}$$

Attachment, $A_{(abc),y} = \langle \hat{a}_c \hat{a}_b \hat{a}_a^\dagger [\hat{H}_A, \hat{a}_y^\dagger] \rangle$ (33 terms, RDM orders [0, 1, 2, 3]):

$$\begin{aligned} & + \delta_{ab} h_{cy} - \delta_{ab} h_{p'y} D_{p',c}^{(1)} - \delta_{ac} h_{by} + \delta_{ac} h_{p'y} D_{p',b}^{(1)} \\ & + h_{by} D_{a,c}^{(1)} - h_{cy} D_{a,b}^{(1)} + h_{p'y} D_{p'a,cb}^{(2)} + \frac{1}{2} \delta_{ab} v_{cy,q's'} D_{q',s'}^{(1)} \\ & - \frac{1}{2} \delta_{ab} v_{cr',q'y} D_{q',r'}^{(1)} - \frac{1}{2} \delta_{ab} v_{p'y,cs'} D_{p',s'}^{(1)} \\ & + \frac{1}{2} \delta_{ab} v_{p'r',cy} D_{p',r'}^{(1)} - \frac{1}{2} \delta_{ab} v_{p'y,q's'} D_{p'q',cs'}^{(2)} \\ & + \frac{1}{2} \delta_{ab} v_{p'r',q'y} D_{p'q',cr'}^{(2)} - \frac{1}{2} \delta_{ac} v_{by,q's'} D_{q',s'}^{(1)} \\ & + \frac{1}{2} \delta_{ac} v_{br',q'y} D_{q',r'}^{(1)} + \frac{1}{2} \delta_{ac} v_{p'y,bs'} D_{p',s'}^{(1)} \\ & - \frac{1}{2} \delta_{ac} v_{p'r',by} D_{p',r'}^{(1)} + \frac{1}{2} \delta_{ac} v_{p'y,q's'} D_{p'q',bs'}^{(2)} \\ & - \frac{1}{2} \delta_{ac} v_{p'r',q'y} D_{p'q',br'}^{(2)} + \frac{1}{2} v_{by,cs'} D_{a,s'}^{(1)} - \frac{1}{2} v_{br',cy} D_{a,r'}^{(1)} \\ & - \frac{1}{2} v_{by,q's'} D_{q'a,cs'}^{(2)} + \frac{1}{2} v_{br',q'y} D_{q'a,cr'}^{(2)} - \frac{1}{2} v_{cy,bs'} D_{a,s'}^{(1)} \\ & + \frac{1}{2} v_{cr',by} D_{a,r'}^{(1)} + \frac{1}{2} v_{cy,q's'} D_{q'a,bs'}^{(2)} - \frac{1}{2} v_{cr',q'y} D_{q'a,br'}^{(2)} \\ & + \frac{1}{2} v_{p'y,bs'} D_{p'a,cs'}^{(2)} - \frac{1}{2} v_{p'r',by} D_{p'a,cr'}^{(2)} - \frac{1}{2} v_{p'y,cs'} D_{p'a,bs'}^{(2)} \\ & + \frac{1}{2} v_{p'r',cy} D_{p'a,br'}^{(2)} - \frac{1}{2} v_{p'y,q's'} D_{p'q'a,cb}^{(3)} + \frac{1}{2} v_{p'r',q'y} D_{p'q'a,cb}^{(3)} \end{aligned}$$

Attachment, $A_{(abc),(y wz)} = \langle \hat{a}_c \hat{a}_b \hat{a}_a^\dagger [\hat{H}_A, \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger] \rangle$ (257 terms, RDM orders [0, 1, 2, 3, 4]):

$$\begin{aligned} & + \delta_{ab} \delta_{wy} h_{cz} - \delta_{ab} \delta_{yz} h_{cw} + \delta_{ab} h_{cw} D_{z,y}^{(1)} - \delta_{ab} h_{cz} D_{w,y}^{(1)} \\ & - \delta_{ab} \delta_{wy} h_{p'z} D_{p',c}^{(1)} + \delta_{ab} \delta_{yz} h_{p'w} D_{p',c}^{(1)} \\ & - \delta_{ab} \delta_{cz} h_{p'w} D_{p',y}^{(1)} + \delta_{ab} h_{p'w} D_{p'z,yc}^{(2)} \\ & + \delta_{ab} \delta_{cw} h_{p'z} D_{p',y}^{(1)} - \delta_{ab} h_{p'z} D_{p'w,yc}^{(2)} - \delta_{ac} \delta_{wy} h_{bz} \\ & + \delta_{ac} \delta_{yz} h_{bw} - \delta_{ac} h_{bw} D_{z,y}^{(1)} + \delta_{ac} h_{bz} D_{w,y}^{(1)} \\ & + \delta_{ac} \delta_{wy} h_{p'z} D_{p',b}^{(1)} - \delta_{ac} \delta_{yz} h_{p'w} D_{p',b}^{(1)} \\ & + \delta_{ac} \delta_{bz} h_{p'w} D_{p',y}^{(1)} - \delta_{ac} h_{p'w} D_{p'z,yb}^{(2)} \\ & - \delta_{ac} \delta_{bw} h_{p'z} D_{p',y}^{(1)} + \delta_{ac} h_{p'z} D_{p'w,yb}^{(2)} + \delta_{wy} h_{bz} D_{a,c}^{(1)} \\ & - \delta_{yz} h_{bw} D_{a,c}^{(1)} + \delta_{cz} h_{bw} D_{a,y}^{(1)} - h_{bw} D_{az,yc}^{(2)} - \delta_{cw} h_{bz} D_{a,y}^{(1)} \\ & + h_{bz} D_{aw,yc}^{(2)} - \delta_{wy} h_{cz} D_{a,b}^{(1)} + \delta_{yz} h_{cw} D_{a,b}^{(1)} - \delta_{bz} h_{cw} D_{a,y}^{(1)} \\ & + h_{cw} D_{az,yb}^{(2)} + \delta_{bw} h_{cz} D_{a,y}^{(1)} - h_{cz} D_{aw,yb}^{(2)} + \delta_{wy} h_{p'z} D_{p'a,cb}^{(2)} \\ & - \delta_{yz} h_{p'w} D_{p'a,cb}^{(2)} - \delta_{bz} h_{p'w} D_{p'a,yc}^{(2)} + \delta_{cz} h_{p'w} D_{p'a,yb}^{(2)} \\ & + h_{p'w} D_{p'az,ycb}^{(3)} + \delta_{bw} h_{p'z} D_{p'a,yc}^{(2)} - \delta_{cw} h_{p'z} D_{p'a,yb}^{(2)} \\ & - h_{p'z} D_{p'aw,ycb}^{(3)} + \frac{1}{2} \delta_{ab} \delta_{wy} v_{cz,q's'} D_{q',s'}^{(1)} \\ & - \frac{1}{2} \delta_{ab} \delta_{wy} v_{cr',q'z} D_{q',r'}^{(1)} - \frac{1}{2} \delta_{ab} \delta_{yz} v_{cw,q's'} D_{q',s'}^{(1)} \\ & + \frac{1}{2} \delta_{ab} v_{cw,q'z} D_{q',y}^{(1)} - \frac{1}{2} \delta_{ab} v_{cw,q's'} D_{q'z,ys'}^{(2)} \\ & + \frac{1}{2} \delta_{ab} \delta_{yz} v_{cr',q'w} D_{q',r'}^{(1)} - \frac{1}{2} \delta_{ab} v_{cz,q'w} D_{q',y}^{(1)} \\ & + \frac{1}{2} \delta_{ab} v_{cr',q'w} D_{q'z,yr'}^{(2)} + \frac{1}{2} \delta_{ab} v_{cz,q's'} D_{q'w,ys'}^{(2)} \\ & - \frac{1}{2} \delta_{ab} v_{cr',q'z} D_{q'w,yr'}^{(2)} - \frac{1}{2} \delta_{ab} \delta_{wy} v_{p'z,cs'} D_{p',s'}^{(1)} \\ & + \frac{1}{2} \delta_{ab} \delta_{wy} v_{p'r',cz} D_{p',r'}^{(1)} + \frac{1}{2} \delta_{ab} \delta_{yz} v_{p'w,cs'} D_{p',s'}^{(1)} \\ & - \frac{1}{2} \delta_{ab} v_{p'w,cz} D_{p',y}^{(1)} + \frac{1}{2} \delta_{ab} v_{p'w,cs'} D_{p'z,ys'}^{(2)} \\ & - \frac{1}{2} \delta_{ab} \delta_{yz} v_{p'r',cw} D_{p',r'}^{(1)} + \frac{1}{2} \delta_{ab} v_{p'z,cw} D_{p',y}^{(1)} \\ & - \frac{1}{2} \delta_{ab} v_{p'r',cw} D_{p'z,yr'}^{(2)} - \frac{1}{2} \delta_{ab} v_{p'z,cs'} D_{p'w,ys'}^{(2)} \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \delta_{ab} v_{p'r', cz} D_{p'w, yr'}^{(2)} - \frac{1}{2} \delta_{ab} \delta_{wy} v_{p'z, q's'} D_{p'q', cs'}^{(2)} \\
& + \frac{1}{2} \delta_{ab} \delta_{wy} v_{p'r', q'z} D_{p'q', cr'}^{(2)} \\
& + \frac{1}{2} \delta_{ab} \delta_{yz} v_{p'w, q's'} D_{p'q', cs'}^{(2)} + \frac{1}{2} \delta_{ab} v_{p'w, q'z} D_{p'q', yc}^{(2)} \\
& - \frac{1}{2} \delta_{ab} \delta_{cz} v_{p'w, q's'} D_{p'q', ys'}^{(2)} - \frac{1}{2} \delta_{ab} v_{p'w, q's'} D_{p'q'z, ycs'}^{(3)} \\
& - \frac{1}{2} \delta_{ab} \delta_{yz} v_{p'r', q'w} D_{p'q', cr'}^{(2)} - \frac{1}{2} \delta_{ab} v_{p'z, q'w} D_{p'q', yc}^{(2)} \\
& + \frac{1}{2} \delta_{ab} \delta_{cz} v_{p'r', q'w} D_{p'q', yr'}^{(2)} + \frac{1}{2} \delta_{ab} v_{p'r', q'w} D_{p'q'z, ycr'}^{(3)} \\
& + \frac{1}{2} \delta_{ab} \delta_{cw} v_{p'z, q's'} D_{p'q', ys'}^{(2)} \\
& - \frac{1}{2} \delta_{ab} \delta_{cw} v_{p'r', q'z} D_{p'q', yr'}^{(2)} + \frac{1}{2} \delta_{ab} v_{p'z, q's'} D_{p'q'w, ycs'}^{(3)} \\
& - \frac{1}{2} \delta_{ab} v_{p'r', q'z} D_{p'q'w, ycr'}^{(3)} - \frac{1}{2} \delta_{ac} \delta_{wy} v_{bz, q's'} D_{q', s'}^{(1)} \\
& + \frac{1}{2} \delta_{ac} \delta_{wy} v_{br', q'z} D_{q', r'}^{(1)} + \frac{1}{2} \delta_{ac} \delta_{yz} v_{bw, q's'} D_{q', s'}^{(1)} \\
& - \frac{1}{2} \delta_{ac} v_{bw, q'z} D_{q', y}^{(1)} + \frac{1}{2} \delta_{ac} v_{bw, q's'} D_{q'z, ys'}^{(2)} \\
& - \frac{1}{2} \delta_{ac} \delta_{yz} v_{br', q'w} D_{q', r'}^{(1)} + \frac{1}{2} \delta_{ac} v_{bz, q'w} D_{q', y}^{(1)} \\
& - \frac{1}{2} \delta_{ac} v_{br', q'w} D_{q'z, yr'}^{(2)} - \frac{1}{2} \delta_{ac} v_{bz, q's'} D_{q'w, ys'}^{(2)} \\
& + \frac{1}{2} \delta_{ac} v_{br', q'z} D_{q'w, yr'}^{(2)} + \frac{1}{2} \delta_{ac} \delta_{wy} v_{p'z, bs'} D_{p', s'}^{(1)} \\
& - \frac{1}{2} \delta_{ac} \delta_{wy} v_{p'r', bz} D_{p', r'}^{(1)} - \frac{1}{2} \delta_{ac} \delta_{yz} v_{p'w, bs'} D_{p', s'}^{(1)} \\
& + \frac{1}{2} \delta_{ac} v_{p'w, bz} D_{p', y}^{(1)} - \frac{1}{2} \delta_{ac} v_{p'w, bs'} D_{p'z, ys'}^{(2)} \\
& + \frac{1}{2} \delta_{ac} \delta_{yz} v_{p'r', bw} D_{p', r'}^{(1)} - \frac{1}{2} \delta_{ac} v_{p'z, bw} D_{p', y}^{(1)} \\
& + \frac{1}{2} \delta_{ac} v_{p'r', bw} D_{p'z, yr'}^{(2)} + \frac{1}{2} \delta_{ac} v_{p'z, bs'} D_{p'w, ys'}^{(2)} \\
& - \frac{1}{2} \delta_{ac} v_{p'r', bz} D_{p'w, yr'}^{(2)} + \frac{1}{2} \delta_{ac} \delta_{wy} v_{p'z, q's'} D_{p'q', bs'}^{(2)} \\
& - \frac{1}{2} \delta_{ac} \delta_{wy} v_{p'r', q'z} D_{p'q', br'}^{(2)} \\
& - \frac{1}{2} \delta_{ac} \delta_{yz} v_{p'w, q's'} D_{p'q', bs'}^{(2)} - \frac{1}{2} \delta_{ac} v_{p'w, q'z} D_{p'q', yb}^{(2)} \\
& + \frac{1}{2} \delta_{ac} \delta_{bz} v_{p'w, q's'} D_{p'q', ys'}^{(2)} + \frac{1}{2} \delta_{ac} v_{p'w, q's'} D_{p'q'z, ybs'}^{(3)} \\
& + \frac{1}{2} \delta_{ac} \delta_{yz} v_{p'r', q'w} D_{p'q', br'}^{(2)} + \frac{1}{2} \delta_{ac} v_{p'z, q'w} D_{p'q', yb}^{(2)} \\
& - \frac{1}{2} \delta_{ac} \delta_{bz} v_{p'r', q'w} D_{p'q', yr'}^{(2)} - \frac{1}{2} \delta_{ac} v_{p'r', q'w} D_{p'q'z, ybr'}^{(3)} \\
& - \frac{1}{2} \delta_{ac} \delta_{bw} v_{p'z, q's'} D_{p'q', ys'}^{(2)} \\
& + \frac{1}{2} \delta_{ac} \delta_{bw} v_{p'r', q'z} D_{p'q', yr'}^{(2)} - \frac{1}{2} \delta_{ac} v_{p'z, q's'} D_{p'q'w, ybs'}^{(3)} \\
& + \frac{1}{2} \delta_{ac} v_{p'r', q'z} D_{p'q'w, ybr'}^{(3)} + \frac{1}{2} \delta_{wy} v_{bz, cs'} D_{a, s'}^{(1)} \\
& - \frac{1}{2} \delta_{wy} v_{br', cz} D_{a, r'}^{(1)} - \frac{1}{2} \delta_{yz} v_{bw, cs'} D_{a, s'}^{(1)} + \frac{1}{2} v_{bw, cz} D_{a, y}^{(1)} \\
& - \frac{1}{2} v_{bw, cs'} D_{az, ys'}^{(2)} + \frac{1}{2} \delta_{yz} v_{br', cw} D_{a, r'}^{(1)} - \frac{1}{2} v_{bz, cw} D_{a, y}^{(1)} \\
& + \frac{1}{2} v_{br', cw} D_{az, yr'}^{(2)} + \frac{1}{2} v_{bz, cs'} D_{aw, ys'}^{(2)} - \frac{1}{2} v_{br', cz} D_{aw, yr'}^{(2)} \\
& - \frac{1}{2} \delta_{wy} v_{bz, q's'} D_{q'a, cs'}^{(2)} + \frac{1}{2} \delta_{wy} v_{br', q'z} D_{q'a, cr'}^{(2)} \\
& + \frac{1}{2} \delta_{yz} v_{bw, q's'} D_{q'a, cs'}^{(2)} + \frac{1}{2} v_{bw, q'z} D_{q'a, yc}^{(2)} \\
& - \frac{1}{2} \delta_{cz} v_{bw, q's'} D_{q'a, ys'}^{(2)} - \frac{1}{2} v_{bw, q's'} D_{q'az, ycs'}^{(3)} \\
& - \frac{1}{2} \delta_{yz} v_{br', q'w} D_{q'a, cr'}^{(2)} - \frac{1}{2} v_{bz, q'w} D_{q'a, yc}^{(2)} \\
& + \frac{1}{2} \delta_{cz} v_{br', q'w} D_{q'a, yr'}^{(2)} + \frac{1}{2} v_{br', q'w} D_{q'az, ycr'}^{(3)} \\
& + \frac{1}{2} \delta_{cw} v_{bz, q's'} D_{q'a, ys'}^{(2)} - \frac{1}{2} \delta_{cw} v_{br', q'z} D_{q'a, yr'}^{(2)} \\
& + \frac{1}{2} v_{bz, q's'} D_{q'aw, ycs'}^{(3)} - \frac{1}{2} v_{br', q'z} D_{q'aw, ycr'}^{(3)} \\
& - \frac{1}{2} \delta_{wy} v_{cz, bs'} D_{a, s'}^{(1)} + \frac{1}{2} \delta_{wy} v_{cr', bz} D_{a, r'}^{(1)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \delta_{yz} v_{cw,bs'} D_{a,s'}^{(1)} - \frac{1}{2} v_{cw,bz} D_{a,y}^{(1)} + \frac{1}{2} v_{cw,bs'} D_{az,ys'}^{(2)} \\
& - \frac{1}{2} \delta_{yz} v_{cr',bw} D_{a,r'}^{(1)} + \frac{1}{2} v_{cz,bw} D_{a,y}^{(1)} - \frac{1}{2} v_{cr',bw} D_{az,yr'}^{(2)} \\
& - \frac{1}{2} v_{cz,bs'} D_{aw,ys'}^{(2)} + \frac{1}{2} v_{cr',bz} D_{aw,yr'}^{(2)} + \frac{1}{2} \delta_{wy} v_{cz,q's'} D_{q'a,bs'}^{(2)} \\
& - \frac{1}{2} \delta_{wy} v_{cr',q'z} D_{q'a,br'}^{(2)} - \frac{1}{2} \delta_{yz} v_{cw,q's'} D_{q'a,bs'}^{(2)} \\
& - \frac{1}{2} v_{cw,q'z} D_{q'a,yb}^{(2)} + \frac{1}{2} \delta_{bz} v_{cw,q's'} D_{q'a,ys'}^{(2)} \\
& + \frac{1}{2} v_{cw,q's'} D_{q'az,ybs'}^{(3)} + \frac{1}{2} \delta_{yz} v_{cr',q'w} D_{q'a,br'}^{(2)} \\
& + \frac{1}{2} v_{cz,q'w} D_{q'a,yb}^{(2)} - \frac{1}{2} \delta_{bz} v_{cr',q'w} D_{q'a,yr'}^{(2)} \\
& - \frac{1}{2} v_{cr',q'w} D_{q'az,ybr'}^{(3)} - \frac{1}{2} \delta_{bw} v_{cz,q's'} D_{q'a,ys'}^{(2)} \\
& + \frac{1}{2} \delta_{bw} v_{cr',q'z} D_{q'a,yr'}^{(2)} - \frac{1}{2} v_{cz,q's'} D_{q'aw,ybs'}^{(3)} \\
& + \frac{1}{2} v_{cr',q'z} D_{q'aw,ybr'}^{(3)} + \frac{1}{2} \delta_{wy} v_{p'z,bs'} D_{p'a,cs'}^{(2)} \\
& - \frac{1}{2} \delta_{wy} v_{p'r',bz} D_{p'a,cr'}^{(2)} - \frac{1}{2} \delta_{yz} v_{p'w,bs'} D_{p'a,cs'}^{(2)} \\
& - \frac{1}{2} v_{p'w,bz} D_{p'a,yc}^{(2)} + \frac{1}{2} \delta_{cz} v_{p'w,bs'} D_{p'a,ys'}^{(2)} \\
& + \frac{1}{2} v_{p'w,bs'} D_{p'az,ycs'}^{(3)} + \frac{1}{2} \delta_{yz} v_{p'r',bw} D_{p'a,cr'}^{(2)} \\
& + \frac{1}{2} v_{p'z,bw} D_{p'a,yc}^{(2)} - \frac{1}{2} \delta_{cz} v_{p'r',bw} D_{p'a,yr'}^{(2)} \\
& - \frac{1}{2} v_{p'r',bw} D_{p'az,ycr'}^{(3)} - \frac{1}{2} \delta_{cw} v_{p'z,bs'} D_{p'a,ys'}^{(2)} \\
& + \frac{1}{2} \delta_{cw} v_{p'r',bz} D_{p'a,yr'}^{(2)} - \frac{1}{2} v_{p'z,bs'} D_{p'aw,ycs'}^{(3)} \\
& + \frac{1}{2} v_{p'r',bz} D_{p'aw,ycr'}^{(3)} - \frac{1}{2} \delta_{wy} v_{p'z,cs'} D_{p'a,bs'}^{(2)} \\
& + \frac{1}{2} \delta_{wy} v_{p'r',cz} D_{p'a,br'}^{(2)} + \frac{1}{2} \delta_{yz} v_{p'w,cs'} D_{p'a,bs'}^{(2)} \\
& + \frac{1}{2} v_{p'w,cz} D_{p'a,yb}^{(2)} - \frac{1}{2} \delta_{bz} v_{p'w,cs'} D_{p'a,ys'}^{(2)} \\
& - \frac{1}{2} v_{p'w,cs'} D_{p'az,ybs'}^{(3)} - \frac{1}{2} \delta_{yz} v_{p'r',cw} D_{p'a,br'}^{(2)} \\
& - \frac{1}{2} v_{p'z,cw} D_{p'a,yb}^{(2)} + \frac{1}{2} \delta_{bz} v_{p'r',cw} D_{p'a,yr'}^{(2)} \\
& + \frac{1}{2} v_{p'r',cw} D_{p'az,ybr'}^{(3)} + \frac{1}{2} \delta_{bw} v_{p'z,cs'} D_{p'a,ys'}^{(2)} \\
& - \frac{1}{2} \delta_{bw} v_{p'r',cz} D_{p'a,yr'}^{(2)} + \frac{1}{2} v_{p'z,cs'} D_{p'aw,ybs'}^{(3)} \\
& - \frac{1}{2} v_{p'r',cz} D_{p'aw,ybr'}^{(3)} - \frac{1}{2} \delta_{wy} v_{p'z,q's'} D_{p'q'a,cb's'}^{(3)} \\
& + \frac{1}{2} \delta_{wy} v_{p'r',q'z} D_{p'q'a,cb'r'}^{(3)} + \frac{1}{2} \delta_{yz} v_{p'w,q's'} D_{p'q'a,cb's'}^{(3)} \\
& - \frac{1}{2} v_{p'w,q'z} D_{p'q'a,ycb}^{(3)} + \frac{1}{2} \delta_{bz} v_{p'w,q's'} D_{p'q'a,ycs'}^{(3)} \\
& - \frac{1}{2} \delta_{cz} v_{p'w,q's'} D_{p'q'a,ybs'}^{(3)} + \frac{1}{2} v_{p'w,q's'} D_{p'q'az,ycb's'}^{(4)} \\
& - \frac{1}{2} \delta_{yz} v_{p'r',q'w} D_{p'q'a,cb'r'}^{(3)} + \frac{1}{2} v_{p'z,q'w} D_{p'q'a,ycb}^{(3)} \\
& - \frac{1}{2} \delta_{bz} v_{p'r',q'w} D_{p'q'a,ycr'}^{(3)} + \frac{1}{2} \delta_{cz} v_{p'r',q'w} D_{p'q'a,ybr'}^{(3)} \\
& - \frac{1}{2} v_{p'r',q'w} D_{p'q'az,ycb'r'}^{(4)} - \frac{1}{2} \delta_{bw} v_{p'z,q's'} D_{p'q'a,ycs'}^{(3)} \\
& + \frac{1}{2} \delta_{bw} v_{p'r',q'z} D_{p'q'a,ycr'}^{(3)} + \frac{1}{2} \delta_{cw} v_{p'z,q's'} D_{p'q'a,ybs'}^{(3)} \\
& - \frac{1}{2} \delta_{cw} v_{p'r',q'z} D_{p'q'a,ybr'}^{(3)} - \frac{1}{2} v_{p'z,q's'} D_{p'q'aw,ycb's'}^{(4)} \\
& + \frac{1}{2} v_{p'r',q'z} D_{p'q'aw,ycb'r'}^{(4)} - \delta_{ab} \delta_{cw} h_{yq'} D_{z,q'}^{(1)} \\
& + \delta_{ab} \delta_{cz} h_{yq'} D_{w,q'}^{(1)} + \delta_{ab} h_{yq'} D_{wz,cq'}^{(2)} \\
& + \delta_{ac} \delta_{bw} h_{yq'} D_{z,q'}^{(1)} - \delta_{ac} \delta_{bz} h_{yq'} D_{w,q'}^{(1)} \\
& - \delta_{ac} h_{yq'} D_{wz,bq'}^{(2)} - \delta_{bw} \delta_{cz} h_{yq'} D_{a,q'}^{(1)} - \delta_{bw} h_{yq'} D_{az,cq'}^{(2)} \\
& + \delta_{bz} \delta_{cw} h_{yq'} D_{a,q'}^{(1)} + \delta_{cw} h_{yq'} D_{az,bq'}^{(2)} + \delta_{bz} h_{yq'} D_{aw,cq'}^{(2)} \\
& - \delta_{cz} h_{yq'} D_{aw,bq'}^{(2)} + h_{yq'} D_{awz,cbq'}^{(3)}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2}\delta_{ab}\delta_{cw}v_{yr',q's'}D_{q'z,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{ab}\delta_{cw}v_{p'r',ys'}D_{p'z,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{ab}\delta_{cz}v_{yr',q's'}D_{q'w,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{ab}\delta_{cz}v_{p'r',ys'}D_{p'w,s'r'}^{(2)} +\frac{1}{2}\delta_{ab}v_{yr',cs'}D_{wz,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{ab}v_{yr',q's'}D_{q'wz,cs'r'}^{(3)} -\frac{1}{2}\delta_{ab}v_{cr',ys'}D_{wz,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{ab}v_{p'r',ys'}D_{p'wz,cs'r'}^{(3)} +\frac{1}{2}\delta_{ac}\delta_{bw}v_{yr',q's'}D_{q'z,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{ac}\delta_{bw}v_{p'r',ys'}D_{p'z,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{ac}\delta_{bz}v_{yr',q's'}D_{q'w,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{ac}\delta_{bz}v_{p'r',ys'}D_{p'w,s'r'}^{(2)} -\frac{1}{2}\delta_{ac}v_{yr',bs'}D_{wz,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{ac}v_{yr',q's'}D_{q'wz,bs'r'}^{(3)} +\frac{1}{2}\delta_{ac}v_{br',ys'}D_{wz,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{ac}v_{p'r',ys'}D_{p'wz,bs'r'}^{(3)} -\frac{1}{2}\delta_{bw}\delta_{cz}v_{yr',q's'}D_{q'a,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{bw}\delta_{cz}v_{p'r',ys'}D_{p'a,s'r'}^{(2)} -\frac{1}{2}\delta_{bw}v_{yr',cs'}D_{az,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{bw}v_{yr',q's'}D_{q'az,cs'r'}^{(3)} +\frac{1}{2}\delta_{bw}v_{cr',ys'}D_{az,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{bw}v_{p'r',ys'}D_{p'az,cs'r'}^{(3)} +\frac{1}{2}\delta_{bz}\delta_{cw}v_{yr',q's'}D_{q'a,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{bz}\delta_{cw}v_{p'r',ys'}D_{p'a,s'r'}^{(2)} +\frac{1}{2}\delta_{cw}v_{yr',bs'}D_{az,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{cw}v_{yr',q's'}D_{q'az,bs'r'}^{(3)} -\frac{1}{2}\delta_{cw}v_{br',ys'}D_{az,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{cw}v_{p'r',ys'}D_{p'az,bs'r'}^{(3)} +\frac{1}{2}\delta_{bz}v_{yr',cs'}D_{aw,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{bz}v_{yr',q's'}D_{q'aw,cs'r'}^{(3)} -\frac{1}{2}\delta_{bz}v_{cr',ys'}D_{aw,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{bz}v_{p'r',ys'}D_{p'aw,cs'r'}^{(3)} -\frac{1}{2}\delta_{cz}v_{yr',bs'}D_{aw,s'r'}^{(2)} \\
& +\frac{1}{2}\delta_{cz}v_{yr',q's'}D_{q'aw,bs'r'}^{(3)} +\frac{1}{2}\delta_{cz}v_{br',ys'}D_{aw,s'r'}^{(2)} \\
& -\frac{1}{2}\delta_{cz}v_{p'r',ys'}D_{p'aw,bs'r'}^{(3)} +\frac{1}{2}v_{yr',bs'}D_{awz,cs'r'}^{(3)} \\
& -\frac{1}{2}v_{yr',cs'}D_{awz,bs'r'}^{(3)} +\frac{1}{2}v_{yr',q's'}D_{q'awz,cb's'r'}^{(4)} \\
& -\frac{1}{2}v_{br',ys'}D_{awz,cs'r'}^{(3)} +\frac{1}{2}v_{cr',ys'}D_{awz,bs'r'}^{(3)} \\
& -\frac{1}{2}v_{p'r',ys'}D_{p'awz,cb's'r'}^{(4)}
\end{aligned}$$

Removal residues, 2h1p rows: $\langle \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_a \hat{a}_x \rangle$ (1 terms, RDM orders [2]):

$$-D_{bc,xa}^{(2)}$$

Removal three-operator amplitudes, 2h1p rows: $\langle \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_a \hat{a}_u^\dagger \hat{a}_v \hat{a}_t \rangle$ ($t = z, u = y, v = w$ of C_3) (2 terms, RDM orders [2, 3]):

$$+\delta_{au}D_{bc,vt}^{(2)} - D_{bcu,vta}^{(3)}$$

Attachment residues, 2p1h rows: $\langle \hat{a}_x \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger \rangle$ (7 terms, RDM orders [0, 1, 2]):

$$\begin{aligned}
& +\delta_{wy}\delta_{xz} - \delta_{wy}D_{z,x}^{(1)} - \delta_{wx}\delta_{yz} + \delta_{wx}D_{z,y}^{(1)} \\
& +\delta_{yz}D_{w,x}^{(1)} - \delta_{xz}D_{w,y}^{(1)} + D_{wz,yx}^{(2)}
\end{aligned}$$

Attachment three-operator amplitudes, 2p1h rows: $\langle \hat{a}_u^\dagger \hat{a}_v \hat{a}_t \hat{a}_y \hat{a}_w^\dagger \hat{a}_z^\dagger \rangle$ (13 terms, RDM orders [1, 2, 3]):

$$\begin{aligned}
& +\delta_{tz}\delta_{wy}D_{u,v}^{(1)} - \delta_{vz}\delta_{wy}D_{u,t}^{(1)} - \delta_{wy}D_{uz,vt}^{(2)} \\
& -\delta_{tw}\delta_{yz}D_{u,v}^{(1)} + \delta_{tw}\delta_{vz}D_{u,y}^{(1)} - \delta_{tw}D_{uz,yv}^{(2)} \\
& +\delta_{vw}\delta_{yz}D_{u,t}^{(1)} - \delta_{tz}\delta_{vw}D_{u,y}^{(1)} + \delta_{vw}D_{uz,yt}^{(2)} \\
& +\delta_{yz}D_{uw,vt}^{(2)} + \delta_{tz}D_{uw,yv}^{(2)} - \delta_{vz}D_{uw,yt}^{(2)} - D_{uwz,yvt}^{(3)}
\end{aligned}$$

ERPA, $\mathcal{M}_{(pq),(rs)} = \langle [\hat{E}_{qp}, [\hat{H}_A, \hat{E}_{rs}]] \rangle$ (natural-orbital basis) (20 terms, RDM orders [1, 2]):

$$\begin{aligned}
& + h_{pr} D_{q,s}^{(1)} - \frac{1}{2} v_{pr,q's'} D_{q'q,ss'}^{(2)} + \frac{1}{2} v_{pr',q'r} D_{q'q,sr'}^{(2)} \\
& + \frac{1}{2} v_{p'r,ps'} D_{p'q,ss'}^{(2)} - \frac{1}{2} v_{p'r',pr} D_{p'q,sr'}^{(2)} - \delta_{pr} h_{sq'} D_{q,q'}^{(1)} \\
& - \frac{1}{2} \delta_{pr} v_{sr',q's'} D_{q'q,s'r'}^{(2)} + \frac{1}{2} \delta_{pr} v_{p'r',ss'} D_{p'q,s'r'}^{(2)} \\
& - \frac{1}{2} v_{sr',ps'} D_{qr,s'r'}^{(2)} + \frac{1}{2} v_{pr',ss'} D_{qr,s'r'}^{(2)} - \delta_{qs} h_{p'r} D_{p',p}^{(1)} \\
& - \frac{1}{2} \delta_{qs} v_{p'r,q's'} D_{p'q',ps'}^{(2)} - \frac{1}{2} v_{p'r,q'q} D_{p'q',sp}^{(2)} \\
& + \frac{1}{2} \delta_{qs} v_{p'r',q'r} D_{p'q',pr'}^{(2)} + \frac{1}{2} v_{p'q,q'r} D_{p'q',sp}^{(2)} + h_{sq} D_{r,p}^{(1)} \\
& - \frac{1}{2} v_{sq,q's'} D_{q'r,ps'}^{(2)} + \frac{1}{2} v_{sr',q'q} D_{q'r,pr'}^{(2)} + \frac{1}{2} v_{p'q,ss'} D_{p'r,ps'}^{(2)} \\
& - \frac{1}{2} v_{p'r',sq} D_{p'r,pr'}^{(2)}
\end{aligned}$$

ERPA, $\mathcal{S}_{(pq),(rs)} = \langle [\hat{E}_{qp}, \hat{E}_{rs}] \rangle$ (2 terms, RDM orders [1]):

$$+ \delta_{pr} D_{q,s}^{(1)} - \delta_{qs} D_{r,p}^{(1)}$$